

Net Gains:

A comparison of competitive balance measures between the ANZ Championship and the Suncorp Super Netball League

A thesis submitted in partial fulfilment of the requirement for the degree of
Bachelor of Business (Economics, Honours)

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18/11/2018

Declaration

I hereby declare that:

- Except where due reference is made in the text of this thesis, to the best of my knowledge and belief, this thesis contains no material previously published or written by another person,
- This thesis contains no material that has been accepted for the award of any other degree or diploma in any university or equivalent institution,
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Abstract

"Fans pay to see a contest, not an exhibition"

-Berri, D., Schmidt, M. and Brook, S. (p.43, 2006)

It is widely believed that competitive balance is critical for the survival of professional athletic leagues. A correlation between "uncertainty of outcome" and fan interest can be detected across multiple geographies and sports, including female netball competitions held in Australia and New Zealand. This thesis will test the levels of competitive balance by calculating measures of concentration, dominance, and in-game equality. These three areas are focussed on intra-season, inter-season, and within-match balance respectively.

To do so it will contrast the success of two female netball competitions - the ANZ Championship, an elite Trans-Tasman female netball competition disbanded in 2016 due to perceived dominance of Australian teams over their New Zealand rivals and its Australian domestic replacement, Suncorp Super Netball, which has enjoyed greater fan attendance, television rights, and player salaries since its inauguration in 2017.

Measuring concentration, dominance, and in-game equality through a variety of standard tests, such as the Herfindahl-Hirschman Index (HHI), Relative Standard Deviations (RSD) and the Average of Real to Idealised Percentage Differences (ARIPD) measure, an original test developed for this thesis, indicated that the Suncorp Super Netball competition generally enjoyed greater competitive balance than its rival. Furthermore, the belief that teams from New Zealand were weaker than their Australian counterparts was supported using the Lifetime Achievement (LAM) and Success Ratio measures. The theory that this was a direct consequence of an uneven playing roster was disproven by developing a hypothetical league and testing for changes in competitive balance.

To test the strength of these results and provide policy recommendations, two hypothetical leagues were created by manipulating the data. By analysing each hypothetical competition - a reimagined ANZ Championship in which each team played each other an even number of times, an ANZ Championship reimagined without the two poorest performing teams and a Suncorp Super Netball League reimagined without the competition's worst team - it was found that there were superior alternatives to improving competitive balance. An ANZ Championship without the two worst teams, or a Suncorp Super Netball competition without Adelaide would have resulted in more competitively balanced competitions.

The increase in concentration after disbanding the ANZ Championship was unexpected, but has important ramifications for the future of women's Netball in Australia. It can guide Netball Australia

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as they compete against other purveyors of professional sport, and may encourage new equalisation measures such as reverse order player drafts, more team revenue sharing, or tighter salary caps. Other sporting leagues find this research of great interest.

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List of Key Terms

ANZ = ANZ Championship

ARIPD = Average of Real to Idealised Percentage Differences

dHHI= Deviated Herfindahl-Hirschman Index

HHI = Herfindahl-Hirschman Index

ISD = Idealised Standard Deviation

LAM = Lifetime Achievement

NAMSIs = National Measure of Seasonal Imbalance

nHHI= Normalised Herfindahl-Hirschman Index

RSD = Relative Standard Deviation

SSN = Suncorp Super Netball

Acknowledgements

Firstly, I would like to thank Mark Stewart for his supervision of this thesis. His thorough editing, passion for the subject, and kindness during the tough times have been critical for my research.

To Deborah and Olivia, thank you for making the family house a home. It is a place where I feel calm and at ease, and has been a major source of joy in 2018.

To Annie and Charlotte, thank you so much for taking the time to read my drafts and suggest improvements. It is a tough task, and your attention to detail has not gone unnoticed.

To RMIT University, and Professor Francis Farrelly in particular, thank you for the understanding with the personal tragedy that struck this year. Your compassion and thoughtfulness has allowed me to continue striving towards my goals.

To Kylene, my beautiful girlfriend, who has stood by me during the long and lonely nights, celebrated the wins and comforted me during the tough times. From the moment you encouraged me to pursue my dreams, to the final submission, you've been a constant source of light.

And finally, I'd like to thank my Dad. His courage to help and support me while battling cancer was a source of endless motivation. While he was unable to see this final piece of work, I know he would have been proud of it. I believe his mantra of simplicity, honesty, and thoroughness echo throughout my words.

1: Introduction

Netball has a long and successful history in Australia at all levels of the game. At grassroots level, netball has more female participants than any other sport (Australian Sports Commission, 2016), while the national team, the Diamonds, have enjoyed remarkable success on the world stage (Australia has won 70% of all international tournaments ever held.) (Netball Australia, 2018). In recent years, female sport has entered the mainstream in Australia, with women's soccer, Australian Rules football and cricket leagues televised nationally. While these leagues survive due to subsidisation from their male equivalents, netball is unique in that it self-supports purely on the strength of a female-only league (ibid).

In 2008, the top five teams from New Zealand and Australian formed the ANZ Championship, which at the time was the world's most elite netball competition. Teams played a 13-week season, with two matches against home country competitors and one against teams from the other country. The league was a success in many ways; commentators claimed that it "lifted professionalism ... in New Zealand and in Australia" (Mitchell, 2016). However, some critics noted the Australian teams were performing significantly better than their New Zealand counterparts. In the nine years of the competition only one New Zealand team, 2012's Waikato Bay of Plenty Magic, were crowned champions. Furthermore, while Australian derby games proved to be popular on television, games between New Zealand teams or between an Australian and New Zealand team struggled for viewers (Mitchell, 2016). The eventual aim was to achieve full professionalisation, i.e. that players and staff make a fulltime living from the game. After years of dominance from the Australian teams and failing to reach beyond semi-professional status, there were calls to disband the ANZ Championship and replace it with two domestic competitions. The hurdle for Netball Australia was that the majority of television revenue came from New Zealand due to their lucrative deal with Sky Sports. In 2016, Netball Australia secured a major television contract that would allow it to run a viable domestic league, and promptly announced that the ANZ Championship would conclude at the end of the year. The Australian replacement, Suncorp Super Netball, and the New Zealand equivalent, the ANZ Premiership, both begin their inaugural seasons in 2017.

The literature suggests that leagues which contain teams of similar ability, or display "competitive balance", are more viable and attract greater fan interest (Neale, 1964). Fans are attracted to unpredictability in results in the short, medium, and long term, and the more evenly balanced a league's teams are, the greater the unpredictability. Professional sporting leagues are like any other type of business; they provide a product (the enjoyment of watching a game) to consumers (fans) at a price (tickets and broadcast deals). The greater the quality of the product, i.e. the more

unpredictable and exciting the games are, the more likely the revenue due to increases in price and consumers. As clubs, players, broadcasters, sponsors, officials and the league itself all benefit from a financially healthy competition, it is in all their best interests to improve competitive balance.

The research question is therefore as follows: Is the Suncorp Super Netball League more competitively balanced than the ANZ Championship? To test this theory, four hypotheses are developed, all of which address this question in different ways. The four hypotheses examine measures of concentration, dominance and in-game evenness, and contrast them for each league. The fourth hypothesis is divided into two parts. The first surveys the comparative strengths of the Australian teams against their New Zealand counterparts. The second part tests whether this was the result of the uneven roster by creating a hypothetical league. A roster, or schedule, is the program of games that each team will play throughout the season, and is usually decided prior to the start of the season.

Measures of concentration examine the measure of closeness between teams within a season, while dominance examines how persistent some teams are at winning over multiple seasons. In-game measures evaluate the closeness of individual matches primarily by considering final margins.

A major difference between measures of concentration and in-game evenness, as compared to dominance, is whether the identities of the teams are relevant. For dominance measures, the identities of the team are critical, whereas for the other two, they are irrelevant.

This thesis is composed of six parts. The first part is the literature review, which briefly summarises the canon of knowledge surrounding the concept of competitive balance in sports leagues. The second part contains the formally posited research question, alongside the four hypotheses. Thirdly, the method is outlined, summarising the various measures that will be used throughout this project. The fourth part contains the results and a discussion on the differences between the two leagues in regards to competitive balance. Two hypothetical leagues are then created by manipulating data, and are used to explore the alternative options Netball Australia could have taken when deciding to disband the ANZ Championship. The final part of the thesis contains the conclusion, bibliography and appendix.

2: Literature review

Rottenberg (1956) applied the concept of competitive balance from economics to sports leagues, and defined it as the evenness of competency between in a league, both in the present and over time. Having teams of comparative competitive ability is considered a necessary condition to maintain economic viability and fan interest (Levin and McDonald, 2009); leagues that are lopsided in regards to talent distribution are considered predictable and fail to capture the imagination of fans (ibid). After all, “people pay to see a contest, not an exhibition” (Berri, D., Schmidt, M. and Brook, S., 2006). The relationship between fan interest and competitive balance is underlined by three assumptions: the first is that fans are attracted to “unexpectedness” (d’Astous and Gagnon, 2007), such as upset or “surprise” victories of weaker teams over their highly favoured opposition. The greater the stratification of talent within a league, the less likely this will occur. Secondly, “indeterminacy in both process and outcome” is an attractive quality for fans (Wertenbroch and Carmon 2006). At the start of each season, there is no way to predict which teams will be crowned champion or relegated, nor is there a way to know how this outcome will be achieved. This uncertainty drives excitement and interest, and thus increases the appeal of the game. Finally, the concept of “pure entertainment value” is posited by York and Miree (2018). They explain that professional sport is a form of entertainment, similar to films or theatre, and that this entertainment is maximised by exciting, high level, “meaningful” games. Leagues which are evenly balanced are more likely to produce these “high-level excitement” match ups, and will thus attract greater fan interest.

While it is clear that having an evenly balanced competition is in the best interests of the league, it is not clear what is in the best interest of each team. All clubs seek two outcomes: to win and make a profit¹. Unless capped, these incentives, would motivate wealthier teams to buy more talent from the available pool, thus increasing their competitive ability. This leads them to win more games, causing them to increase their supporter base and thus revenue. If this additional revenue is invested into purchasing more playing talent, the cycle has repeated itself. This process, left unabated, causes weaker teams to lose market share to the point where they would be forced to exit the league, ultimately shrinking the league until only the most dominant team is left (Kessene, 2006; Zimbalist, 2003). Professional sport cannot support a monopoly; even the most adored teams require at least one opponent for them to provide a product.

¹ In Australia and Europe, it can be assumed that most clubs are trying to maximise wins and on-field success. In the United States, clubs are more frequently operated as for-profit franchises.

While traditional firms and industries dislike competition, sporting teams require it. As explained by Rottenberg (1956, p.254) “no team can be successful unless its competitors also survive and prosper sufficiently so that the difference in the quality of play among teams are not too great”. This “difference” was further examined by Jones (1969, p.3) who described it as “competitive equality”. The more common term is competitive balance, which is driven by “uncertainty of outcome”, a term used by Quirk and Fort (1992). Syzmanski (2003) defined three forms of uncertainties: championship uncertainty, seasonal uncertainty (i.e. who qualifies for finals or playoffs), and match uncertainty. The hypotheses posited in this thesis are derived from these three definitions.

Failure to preserve competitive balance results in a “Tragedy of the Commons” situation. This phenomenon that was first observed by Garrett Hardin (1968), who used it to explain the behaviours of farmers using common grazing areas. He posited that a situation in which all farmers maximised their use of the common grazing area, would result in insufficient grass to sustain their herds, resulting in starvation and the farmers being ultimately worse off. This is an analogy that explains how individual efforts to self-maximise may cause the group, and ultimately the individual, to be worse off in the long run. Other examples are national parks with too many visitors, or oceans that are overfished (ibid).

As “each league competitor has an inherent interest in maintaining the health of their rivals” (Groot, 2008, p.25), some leagues enforce rules and regulations to create a more even competition. Some are off-field, such as revenue sharing, which forces the wealthier teams to distribute some of their revenue to their poorer rivals. There is significant debate in literature as to whether this aids competitive balance; Fort and Quirk (1995) argue that it has no effect in aiding competitive balance. This has been challenged by Vrooman (1995), Marburger (1997) and Kesenne (2000), although all three have varied the assumptions made in the original Fort and Quirk model. The other mechanisms that can be used to balance leagues are those that equalise talent dispersion. These include salary caps on player remuneration and player drafts, which stop wealthy teams acquiring all the top talent and distributes better players to weaker teams. Salary caps put a maximum limit on how much teams can remunerate their players, thus preventing the richest teams procuring all the top players. Player drafts are a mechanism which allows the worst team from the previous season to have first pick of the available players in the subsequent season. The second worst team picks second, and so on until the best team has a pick, and then the pattern repeats itself. Daly and Moore (1981) found both of these measure to be effective at creating a more even league.

The list of leagues the literature has analysed extensively is large, albeit focused. The majority of research done on “open” leagues, that is, leagues that practice promotion and relegation, is focused

on the “Big 5”, which are Europe’s five biggest soccer leagues. They are: England’s Premier League, France’s Ligue 1, Germany’s Bundesliga, Italy’s Serie A and Spain’s La Liga. In more recent years, there has been analysis conducted on the Austrian and Swiss Leagues (Pawlowski, Nalbantis and Coates, 2015) as well as lower leagues within England (Wilson, Ramchandani and Plumley, 2017).

Studies of closed leagues are mostly focussed on the four major American sporting codes; basketball (NBA), baseball (MLB), American football (NFL) and ice-hockey (NHL) (Mills and Fort, 2014). Apart from having the same teams year after year², these competitions also have a number of equalisation mechanisms such as drafts and salary caps that are rarely used in European soccer. Furthermore, US clubs are far more profit-orientated than their European counterparts. Other researchers have examined competitive balance in Australia (Booth, 2004) and basketball in Europe (Williams, 2012), although it is not as extensive. More recently, the literature has begun to explore competitions such as tennis (Corral, 2009) and Formula One (Schreyer and Torgler, 2016), although due to their non-league and in some cases non-match structure, these studies are not relevant for this thesis.

The literature indicates the idea that closed leagues are more competitive than open leagues in both concentration and dominance (Buzzacchi, Szymanski and Valletti, 2003). As an extreme example of dominance, the Scottish Premier League and German Bundesliga have both had occasions where one team has won the title for six consecutive years, whereas the NFL has never had a single team win three times consecutively.

Though it is widely recognised, the idea of competitive balance is still being developed and critiqued. Pawlowski (2013) believes that the empirical evidence does not correspond well with the established proxies, and this can lead to flawed conclusions. Furthermore, there are a number of studies that argue that competitive balance, while important, is overstated in terms of fan engagement (Andreff and Scelles, 2015; Scelles et al., 2013). This area of research is outside the scope of this thesis.

Regardless of the arguments within academic literature, leagues around the world express the view that competitive balance is important. They act on this view by implementing policies that aim to achieve more balanced competitions; whether that be through revenue-sharing in the Premier League, reverse-order drafts in the AFL or salary caps in Major League Baseball. Despite the importance of competitive balance for the various stakeholders involved in trans-Tasman netball, there has been no formal research in this area, thus justifying this thesis as one of academic integrity and relevance.

² From time to time leagues will expand and accept new teams into their competition, although this is relatively rare. Across the four major American sports, there have only been five new teams added since 2000.

3: Research Question

The proposed question for this thesis is as follows:

“Is the Suncorp Super Netball League more competitively balanced than the ANZ Championship?”

To answer this, four hypotheses are tested, reflecting the three areas of competitive balance, as well as the idea of geographically based inequality.

4: Hypotheses

Hypothesis One: The Suncorp Super Netball league is less concentrated than the ANZ Championship.

Hypothesis Two: The Suncorp Super Netball league exhibits less dominance than the ANZ Championship.

Hypothesis Three: The Suncorp Super Netball league has games which are more competitive than the ANZ Championship.

Hypothesis Four.A: The New Zealand teams were weaker than the Australian teams when measured head to head (ANZ Championship only).

Hypothesis Four.B: The uneven playing schedule did not affect the differences in strength between Australian and New Zealand teams (ANZ Championship Only).

4.1: Explanation of key terms

Zimbalist (2002) defines the difference between concentration and dominance as follows:

- Concentration is the extent of the closeness between teams in a league in a season
- Dominance is the extent to which the same teams persist in winning over a number of seasons.

The key difference between these two concepts is whether the identity of the team is relevant. In measures of concentration it does not matter, whereas for measures of dominance it does. Almost every year in the AFL, there are multiple teams that can qualify for the finals in the last round of the season, suggesting a league with a high level of concentration. In the Bundesliga, the top tier German soccer league, Bayern Munich, has won the championship 13 times in the past two decades, suggesting a very dominant team. In-game measures examine how close and competitive the games are on the assumption that the closer the result, the more balanced the contest.

5: Method

This thesis analyses the competitive balance of both the Suncorp Super Netball and the ANZ Championship using a number of measures for concentration, dominance, and in-game competitiveness. Once analysed, the results for each measure are compared against one another to reveal the difference, if any, between the competitive balance of each league. Zimbalist (2002, p. 112) notes that “[t]here are almost as many ways to measure competitive balance as there are to quantify the money supply”, however, using all the available measures is neither practical nor realistic. Due to the paucity of data owing to the short lifetime of the Suncorp Super Netball league, many of the measures, particularly for dominance, are not suitable. For example, a common test for dominance is consecutive title wins. With only two seasons of the Suncorp Super Netball league, this measure is unlikely to be very illuminating. As such, the measures iterated in Table One are used. A brief summary of what they measure, as well as their strengths and weaknesses, is included.

Due to their number, readers may find it difficult to keep track of the measures used and their relevance to competitive balance. As an aid, two supplementary tables are included to provide a simple reference to assist the reader. It is recommended that they be either printed off separately or removed from the hard copy, and used alongside while reading through this document. Both tables can be found in section 12 at the very end of the thesis.

Table 1: Summary of measures used, their strengths and weaknesses

Name of Measure	Type of Measure	Summary	Strengths	Weaknesses
Herfindahl-Hirschman Index	Concentration	Measures clubs' share of wins	Simple to calculate. Well supported in the literature	Differences between sporting leagues and industry
Relative Standard Deviation	Concentration	Comparison between actual and ideal standard deviations	Can be used for leagues of different sizes	Upper bounds depend on team numbers
National Measure of Seasonal Imbalance	Concentration	Normalisation of Standard Deviations	Can be used for leagues of different sizes	Computationally complex
Concentration Ratios	Concentration	Measures subset against a broader group	Used in a variety of different sports	Weaker results when a league changes in size
Surprise Index	Concentration	Measure using upset victories in a given season	Can be widely used as a comparison tool	Laborious calculation process
Lifetime Achievement	Dominance	Measure of how many standard deviations away the W/L record is from the "ideal"	Can be used irrespective of sport, league, number of games played and time	Becomes more accurate with more data.
Time Series Associations	Dominance	Measures the continuity of membership within a subset from year to year	Simple to calculate	Requires at least two years of data. Subset size is arbitrary
Herfindahl-Hirschman Index	Dominance	Measures club's proportional share of dominance	Simple to calculate. Well supported in the literature	Lack of seasons limits available data
Average Margins	In Game Competitiveness	Average margin of games	Simple to calculate	"The scoreboard didn't tell the whole story"
"Blow-Out" Index	In Game Competitiveness	% of games with margins > K	Can be widely used as a comparison tool	K value is arbitrary
Average of Real to Idealised Percentage Differences	In Game Competitiveness	The average of the absolute differences of actual percentages to the ideal percentage	Can be used for leagues of different sizes, and also to different sports	Original calculation developed for this thesis, and is thus unsupported in academic literature
League Score Average	Concentration	Summing the league rankings	Simple to calculate	Does not account for other factors

5.1: Measures of Concentration

Measurements of concentration focus on the closeness of a league competition, as compared to the relative performances of teams within that league (Evans, 2014). Theoretically, the most concentrated possible league would be one which included only two teams, with one team winning all the games. By definition, the wins are all concentrated in one club. A very unconcentrated league would be one where all teams win 50% of their games; for example, a hypothetical coin flipping league. *Ceteris Paribus*, leagues that are less concentrated have more competitive balance.

In his review of competitive balance measures, Evans (2014) found eight measures that can be used to determine concentration. They are:

- Range
- Excess tail frequencies
- Standard Deviations
- HHI
- Concentration Ratios
- Index of dissimilarity
- Lorenz curve and Gini coefficient
- Surprise index.

However, due to the scope of this study, the four measures tested are:

- HHI
- Standard Deviation, including ratio of “idealised” Standard Deviation
- Concentration ratios, including “Five Club” ratio and “Attainable Points” ratio
- Surprise Index.

They have been chosen due to their robust support in the relevant literature and their applicability to the hypotheses proposed. A brief summary of each measure is presented below:

5.1.1: Herfindahl-Hirschman Index

$$HHI = \sum_{i=1}^N s_i^2$$

Where:

s= Market Share of firm i

The HHI has traditionally been used to measure concentrations of market share within industry by summing the squared percentages of a firm's individual market share. HHI has a maximum score of 1, i.e. an industry with a complete monopoly, and a minimum of $1/N$, which indicates perfect competition. This measure was adapted to sports economics by Depken in 1999 by replacing market share with share of either a team's points or wins, and the term "industry" with "league". The HHI for sports is expressed formally as such:

$$HHI^C = \sum_{i=1}^N \left(\frac{W_i}{\sum_{i=1}^N W_i} \right)^2$$

Where:

W_i = Number of wins for team i

N = Number of teams in the league

To gain a greater clarity of insight into competitive balance, Depken (1999) developed the deviated Herfindahl-Hirschman Index (dHHI) by subtracting the minimum value ($\frac{1}{N}$) from the actual value. The greater this value, the more unbalanced a league was, whereas a minimum score of zero indicated high levels of balance. While this development was useful for comparing leagues of the same size, i.e. from year to year, it made comparisons between leagues with differing numbers of teams difficult.

$$dHHI = \sum_{i=1}^N \left(\frac{W_i}{\sum_{i=1}^N W_i} \right)^2 - \frac{1}{N}$$

In 2007, Owen and King (2007) solved this issue by normalising the dHHI i.e. dividing it by the $dHHI_{\text{Max}}$ value, to produce the Normalised Herfindahl-Hirschman Index (nHHI). nHHI has a minimum score of zero (perfect balance), and a maximum score of one (perfectly imbalanced). This measure will be the one used to compare the ANZ Championship with the Suncorp Super Netball League, by comparing the averages of both leagues. Due to normalisation, the difference of league size and composition can be accounted for.

The nHHI is defined as such:

$$nHHI = \frac{dHHI}{dHHI_{max}}$$
$$dHHI = \sum_{i=1}^N \left(\frac{W_i}{\sum_{i=1}^N W_i} \right)^2 - \frac{1}{N}$$
$$dHHI_{max} = \frac{N+1}{3N(N-1)}$$

Where:

P_i = Team i 's share of all games (or points) won in a season

N = Number of teams in the league

5.1.2: Standard Deviations

Normal Standard Deviation

Standard Deviation is a statistical measure of dispersion from the mean. It is applied in many fields to determine dispersion, and in sports economics as a measure of concentration. In its general sense, Standard Deviation (σ) is calculated as such:

$$\sigma = \sqrt{\frac{\sum_{i=1}^N \left(X_i^* - \frac{\sum_{i=1}^N X_i^*}{N} \right)^2}{N}}$$

Where:

X_i^* = Selected Variable, i.e. games won, points attained etc.

N = number of teams (i) in the league

Idealised Standard Deviation

In a perfectly competitive league, each team would win exactly 50% of their games, which can be defined as the “ideal” situation, assuming the league is trying to optimise “competitive balance”. The actual results often deviate from this “ideal” scenario, and by dividing the actual standard deviation by the ideal standard deviation, a ratio between these two values can be calculated to indicate how balanced a league is in terms of concentration. The closer the ratio is to one, the closer the league is to the perfectly balanced “ideal”.

Quirk and Fort (1992) originally proposed a binomial Idealised Standard Deviation (ISD), expressed formally as $\sigma_I^{(w,l)} = \frac{0.5}{\sqrt{G}}$ where G=number of games per team. This calculation is, however, too simplistic as it ignores draws completely. While this measure may be suitable for games where draws are rare or impossible, for the purposes of this report it is unsuitable. The approach that Cain and Haddock (2006) propose rejects the nested logic that the probability of a draw is zero, and instead incorporates it into their measurement. Expressed formally, the ISD as a share of maximum wins is:

$$\sigma_I^{(\alpha, 0.5\alpha, 0)} = \sqrt{\left(\frac{(1-d)}{4 \times G}\right)} \quad (8)$$

d = percentage chance of a draw.

G= Number of games played

(1-d) = Chance of a win = Chance of a loss.

The superscript, $\alpha, 0.5\alpha, 0$ refers to point allocation. The ANZ Championship distributes points on a 2,1,0 basis, whereas the Suncorp Super Netball league uses a 4,2,0 system. Both are compatible with this measure of ISD. The alternative, 3,1,0, is commonly used in European soccer.

5.1.3: National Measure of Seasonal Imbalance (NAMSI)

Standard deviations can be normalised so as to return values with upper and lower bounds between one and zero respectively. In 2006, Goosens developed a measure known as NAMSI, with zero reflecting a perfectly competitive league and one being perfectly uncompetitive. The benefit of having a normalised measure of standard deviation is that it can be used to directly compare leagues of different sizes. As the Suncorp Super Netball league had a contraction in the number of teams compared to the ANZ Championship, NAMSI will be the more valid measure when comparing the two leagues.

NAMSI is calculated formally as:

$$NAMSI = \sqrt{\frac{\sum_{i=1}^n (w_i - 0.5)^2}{\sum_{i=1}^n (w_{i \max} - 0.5)^2}}$$

Where:

W_i = Win percentage of team i

$W_{i \max}$ = Maximum value of win percentage of team i

N = Number of teams in the league

5.1.4: Concentration ratios

The concentration ratio measures the concentration of a subset of the teams in the league (i.e. a number of top teams) relative to a comparative statistic (Evans, 2014, p.26). There are two variations of this measure: the attainable concentration ratio, which was devised by Koning in 1999, and the Five Club concentration ratio (Michie and Oughton, 2004).

Attainable concentration ratio

The attainable concentration ratio measures the number of points acquired by the top k teams in a league, divided by the maximum possible amount of points that this subset could attain. This measure has a maximum value of one, which represents a league with the least possible balance. Concentration measures the evenness of teams within a league, and one which has the top k clubs acquiring all possible points indicates that the competition is unbalanced.

It is expressed as an equation below:

$$C_k = \frac{\sum_{k=1}^K P_k}{KW(2N - K - 1)}$$

Where:

K= the number of top teams

W= the number of points awarded for a game won

P_k = the number of points won by the kth from top team

N= Number of teams in the league

The minimum value is calculated using the following formula:

$$C_K^{min} = \frac{D(2N-2)}{W(2N-1-K)}$$

K= the number of top teams

W= the number of points awarded for a game won

D= the number of points awarded for a drawn game

N= Number of teams in the league

The value of K is arbitrary, although a general rule is that the larger the league, the larger the k value can be. For the purpose of this thesis, this measure is calculated twice, firstly with k=1 and then again with k = 4, as the top four teams during the regular season qualify for the finals. Koning did the same for the elite Dutch soccer league from 1955/56 to 1998/1999

Five Club Concentration ratio

An alternative to the Koning “attainable concentration ratio” is the Five Club Concentration ratio, henceforth referred to as the C5 ratio. Expressed informally, the C5 ratio is the total points won by the top five clubs divided by the total number of points won by all clubs. This value was converted into an index by Michie and Oughton in 2004, which was based on the perfectly competitive “ideal” minimum value of the measure. This is calculated as:

$$C5R = \frac{\sum_{i=1}^5 (N - X_i)}{\sum_{i=1}^N (N - X_i)}$$

Where:

N = Number of teams in the league

X_i= League position of team i with the top club ranked 1 and the bottom club ranked N

$$C5ICB = \left(\frac{C5\ ratio}{5/N} \right) \times 100(13)$$

Where:

N= Number of teams in the league

This index approach is considered the more intuitive interpretation, and allows leagues of different sizes to be compared against each other. However, the upper limits of the measure still depend on the point system and the number of teams in the competition.

5.1.5: Surprise Index

One of the thrills of competitive sport is when an underdog beats a more favoured opponent (d'Astous and Gagnon, 2007). Competitions such as the FA Cup, a knockout soccer tournament in England, markets itself as much on “miracle” wins as it does on clashes between the Premier League heavyweights. Groot and Groot (2003) used this idea to create a surprise index which expanded this idea to help calculate concentration within a league. A “surprise” is defined as a win or draw over a team that finishes the season higher on the table. This value is then weighted by the difference between the two teams to reflect the magnitude of surprise, and then divided by the number of surprise points within a perfectly balanced league. A perfectly balanced league has a value of one, whereas one which has a score of zero is completely unbalanced (Evans, 2014).

The Surprise index is calculated as:

$$S_l = \frac{\sum_{i=1}^{N-1} \sum_{j=i+1}^N (R_{ij} + R_{ji})(j - i)}{\frac{N(N+1)(N-1)}{3}} \quad (14)$$

Team i ends the season in a higher position in the league table (i.e. with a lower rank number) than j ($i < j$). It should be noted that the rankings are static, and are taken from the table at the end of the season.

R_{ij} and R_{ji} = The result of the game between team i and j. The order of the i and j refers to whether the teams play home or away, with the first letter denoting the home team.

Scoring:

$R_{ij}, R_{ji} = 2$ if the game results in a win for team j

$R_{ij}, R_{ji} = 1$ if the game results in a draw

$R_{ij}, R_{ji} = 0$ if the game results in a win for team i

N = Number of teams in the league

These scores are then multiplied by the difference between the two teams' ranks to give the number of “surprise points”. The sum of these values is then divided by the total number of surprise points in a perfectly balanced league ($\frac{N(N+1)(N-1)}{3}$) to provide the Surprise Index.

As an example to aid understanding, consider a league featuring teams A, B, and C, which at the end of the season finish first, second, and third respectively. Assume that during the season, team C beat team A, and also drew with team B. The win against team A is considered a surprise, so it is worth two points, and it is then multiplied by two, the difference between team A's final ranking (first) and team C (third). This equals four surprise points in total. Additionally, the draw between team C and B is a surprise, and is awarded one point for the draw, and one point for the gap between the two teams, equally a total surprise points score of one. Assuming there are no other surprises in the season, the sum of surprise points is $(2 \times 2) + (1 \times 1) = 5$. To calculate the surprise index, divide by the denominator, which in this instance is 8. Therefore the surprise is $\frac{5}{8} = 0.625$.

An advantage of this measure is that it is very adaptable; it can be used to compare leagues of different sizes, seasons and even sports. While some critics argue that it requires a more laborious approach to data collection, Groot and Groot argue that all the information is readily available (2003).

5.2: Measures of dominance

Dominance measures the mobility of teams within a league in regards to success. A league with high dominance will see a small number or even a single team win consistently year after year, while the remaining teams are confined to mediocrity. Unlike concentration, the identities of the club are essential for the calculations. There are seven widely accepted measures for dominance, which are:

- Descriptive Statistics
- Time Series Association
- Lifetime Achievement Measure (LAM)
- Herfinahl-Hirschman Index (HHI)
- Lorenz Curve
- Gini Coefficient
- Surprise Index for teams.

Descriptive statistics include such measures as “consecutive title wins” and “frequency of failure to win a title”. The Lorenz Curve and Gini Coefficient are both unsuitable for this thesis, and are thus excluded from it. Furthermore, the Surprise Index for dominance does not address any of the four hypotheses posited, and is thus also disregarded.

5.2.1: Time Series Association

Time series measures compare two or more consecutive seasons against each to determine what changes have occurred. There are three variations on how to do this: correlation coefficient, adjusted churn, and “top 4” reoccurrence. The correlation coefficient measures the relationship one set of data to another set, with values ranging between zero (no correlation) to one (complete correlation).

Correlation Coefficient

The formal expression of the correlation coefficient is:

$$r = \frac{\sum_{i=1}^N (x_i y_i) - \bar{X} \sum_{i=1}^N y_i}{\sqrt{[\sum_{i=1}^N (x_i^2) - \bar{X} \sum_{i=1}^N x_i][\sum_{i=1}^N (y_i^2) - \bar{Y} \sum_{i=1}^N y_i]}}$$

Where:

N = Number of teams (i) in the league

$\bar{X}, \bar{Y}$ = Average of values for the periods x and y respectively

x_i = Win percentage of team i in period x

y_i = Win percentage of team i in period y

Excel spreadsheets allows one to calculate the correlation coefficient quickly and simply, and it is this method that is used for this thesis.

Adjusted Churn

Adjusted Churn was proposed by Mizak, Neral and Stair in 2007 to calculate the absolute difference in league rank for clubs in consecutive years. The formula is expressed as:

$$AC_t = \frac{C_t}{C_{t \max}}$$

Where:

C_t = Churn for season t

$$C_t = \frac{\sum_{i=1}^N |k_{i,t} - k_{i,t-1}|}{N}$$

$C_{t \max}$ = The maximum possible value of $C_t = N/2$

$K_{i,t}$ = Rank of team i in season t

N = number of teams in the league

A league which has a complete reversal of rankings from one season to the next would have an adjusted churn score of ones, indicating perfect balance (Mizak, Neral and Stair, 2007). The minimum value of zero reflects a league that has completely ossified rankings from the previous year.

Top Four Reoccurrence

This measure, proposed by Curren, Jennings, and Sedgwick in 2009, calculates the probability of the top four teams repeating the performance in the preceding year. If all four teams stay within the subset from year one to year two, the value is 100%, regardless of internal ranking within the subset. A score of 0% means that no team from the initial top four were able to repeat their success. It is calculated as:

Table 2: Top Four Reoccurrence

Number of teams which remained in top four from previous season	Percentage Reoccurrence
No top four teams from previous season were in top four of current season	0%
One reoccurring team, three new teams	25%
Two reoccurring teams, two new teams	50%
Three reoccurring teams, one new team	75%
Complete reoccurrence from previous season of top four places	100%

5.2.2: Herfindahl-Hirschman Index

The HHI has been explained previously in this proposal as a measure for concentration. It can also be used to measure dominance in two ways; the number of titles won or the number of teams within a top position. In the same way the HHI was used for concentration, it has a maximum value of one and a minimum of $\frac{1}{N}$.

The formula used to express this formally is:

$$HHI^D = \sum_{i=1}^N x_i^2 \quad (17)$$

Where:

x_i = Share of (insert determinant here, i.e. top 5, champion etc.) of team i over a selected number of seasons

N = Number of teams in the league.

5.2.3: Lifetime Achievement

This measure uses a team's average win percentage over its "lifetime" in a league as a basis to measure how many standard deviations away from the "ideal" winning percentage it has deviated. The ideal winning percentage is defined as 0.5, i.e. a team wins exactly half of the games played. The benefits of this calculation are myriad; it is not dependent on league size or composition, or even sport. It is therefore possible to compare the ANZ's Canterbury Tactix with Super Netball's Sunshine Coast Lightning, and even against American baseball's New York Yankees. By determining which teams are over-performers, under-performers and the range between them, it is possible to infer great insights into the competitive balance of a league.

$$Lifetime\ Achievement = \frac{\bar{w}_i - 0.5}{\bar{\sigma}_{ii}}$$

$w_{i,t}$ =Win percentage of team i for year t in the league

T_i =Total number of years in the league for team

$\bar{\sigma}_{ii}$ =The idealised standard deviation for team i= $\frac{0.5}{\sqrt{G} \times \sqrt{T}}$

5.3: Competitive Balance expressed in game evenness measures

Due to the scarcity of seasons for analysis, this study examines the evenness of individual games to determine whether competitive balance has improved. There are three such tests: the first focuses on descriptive statistics regarding game margins, the second is the "Blow-Out" index, and the third is the Average of Real to Ideal Percentage Differences (ARIPD). To the best of the author's knowledge, ARIPD is an original measure designed specifically for this thesis.

5.3.1: Average margins of games

Competitive Balance can be measured by the margin of victory between the winning and losing team for each game (Carroll and Humphrey 2016, Sutter and Winkler, 2003; Paul, Wachsman and Weinbach 2011). This measure assumes that the smaller the margin, the greater the “outcome uncertainty” is, and therefore the greater the competitive balance. To test whether or not game margins have decreased from the ANZ Championship to Suncorp Super Netball, the margins of each game will be taken, averaged and then compared. If the average margins decrease, then there is evidence that games are becoming more competitive, and thus the league has more competitive balance.

$$\text{Average Margin} = \frac{\sum_G (P_w - P_l)}{G} \quad (20)$$

Where:

P_w = Points scored by winning team

P_l = Points scored by losing team

G = Total games played

5.3.2: “Blow-Out” index

A “Blow-Out” is a colloquial term used to describe a game where the final margin is considerable. As much of the excitement and pleasure from professional sport comes from the “uncertainty of outcome” (Rottenberg, 1956), large margins, and thus very certain outcomes, are counterproductive in terms of sustaining fan interest in the game. The “Blow-Out” index is a measure of what percentage of a season’s games end in “Blow-Outs”, with a “Blow-Out” defined as a game with a margin over k points. This value can be compared between seasons or between leagues, regardless of league size or length. The point k is arbitrary, but for the purpose of this study it will be defined as ten. This was calculated by finding the average score per team per game for the lifetime of the ANZ Championship, which is 54.55. If games that have a final margin of 20% or more of the average score are classified as blowouts, then the k value is 10.91 (20% of 54.55). As margins can only be expressed as integers, the value of k is rounded down to ten. To compare leagues of different sizes and lengths, the measure can be converted to a percentage by dividing the number of “Blow-Outs” by the total number of games played.

5.3.3 Average of Real to Idealised Percentage Differences

Introduction to percentages

A problem that many sporting leagues have is how to rank teams who have recorded an identical number of points. There are a number of possible solutions used around the world, in European soccer the convention is to use goal difference. An alternative to goal difference is a goal ratio, commonly referred to as the “percentage”. The goal ratio is the number of goals or points scored, divided by the number of goals or points conceded, multiplied by 100. A team that has scored 20 goals over the season, and conceded ten, will have a “percentage” of 200%. This system is used in both the ANZ Championship and the Suncorp Super Netball league, as well as other sports, such as the AFL.

Explanation of Average to Real to Ideal Percentage Differences

The ARIPD is an original calculation developed specifically for this thesis that builds upon the principles of Relative Standard Deviations to provide a more complete understanding of competitive balance. RSD is less precise than ARIPD, as it treats all wins the same, regardless of whether the final margin is one point or a hundred points. By using percentages as an input, ARIPD is capable of delivering deeper and more precise insights. To the best of the author’s knowledge, this is a new contribution to field of Sports Economics.

The ARIPD is calculated as:

$$ARIPD = \frac{\sum_{i=1}^N |P_i - 1|}{N} \quad (21)$$

N= Number of teams in league

P₁ = Percentage of team i

The ARIPD measure is the averaged deviation from the idealised mean of 100% (expressed as 1 as an integer). A score of zero is “perfectly competitive”, as each team would have a realised percentage of exactly 100%. A value of 100% means that every time a team scored a point, a point was scored against it in the season. Another way of thinking about it is that if the season was just one long game, the percentage is the final margin. If competitive balance is the objective, this is the ideal percentage. This calculation makes the assumption that all teams are trying to maximise their scores at all times, which may be unrealistic. For example, it is not uncommon for soccer teams to bench their star players when playing against weaker opposition as a tactic to reduce possible injuries or give experience to younger athletes.

The minimum value of ARIPD is zero. This would indicate that every team finished the season with a percentage of 100, and thus the league is entirely balanced. The upper bound is infinite. The higher the ARIPD value, the more unbalanced the league is.

To further explain how the ARIPD is calculated, an example is used.

Imagine a league with three teams, the results of which can be seen in table three:

Table 3: ARIPD Example League

Team Name	Points for	Points Against	Percentage	ARIPD Value
A	20	10	200%	1
B	15	15	100%	0
C	10	20	50%	0.5

The ARIPD measure would calculate the absolute difference between the actual percentages of teams A, B, and C, and the ideal percentage of 100%. The value needs to be absolute to prevent negative values from cancelling out positive values. The sum of the individual ARIPD values is 1.5, and by dividing it by the number of teams, three, the ARIPD index can be found. In this example, it is 0.5.

5.3.4: Benefits of ARIPD

There are multiple benefits of ARIPD. Firstly, it takes into account all the teams, as compared to the range which only reflects the difference between the best and worst teams. A league where all teams are close to the ideal, bar one highly dominant team, would have a lower ARIPD score than one with the same range, but more evenly spread out. This can be seen in the extreme, hypothetical example outlined in table four. While both leagues have a range of 100, their league structure is very different. League A has one highly dominant team, and nine identically balanced teams. League B is more traditional, with each team ranked in ten percentage point intervals. The question is, which league is superior? It all depends on what the organisers consider to be more important when it comes to uncertainty. If the goal is to maximise the uncertainty of who will be crowned champion, then League B is significantly better, as Team One in League A is overly dominant compared to its opposition. However, if the organisers and fans want to watch and attend close games, then league A is superior. Unless playing against team one, all teams in League A will have an equal chance of winning, as their talents are identical, thus the likelihood of having close games is maximised.

Critics may argue that this may be counteracted by the lack of interest in games that Team One play, considering how one-sided the results would probably be. However, research shows that the fans of

weak teams are not necessarily put off when playing against top teams. In fact, fan attendance often increases, as fans are attracted to witnessing top players (the Superstar Effect) or the possibility of seeing a historic underdog victory (Berri and Schmidt, 2006). Both of these factors indicate that League A is superior to league B in a “within-season” and “within-game” context, a result confirmed by the derived ARIPD measures.

Table 4: Example ARIPD Table

	League 1	League 2
Team A	190	150
Team B	90	140
Team C	90	130
Team D	90	120
Team E	90	110
Team F	90	90
Team G	90	80
Team H	90	70
Team I	90	60
Team J	90	50
Std. Dev	30	33.17
Min	90	50
Max	190	150
Range	100	100
ARIPD	18	30

The second benefit of the ARIPD score is that it can be used to compare leagues of different sizes in terms of both games played and teams involved. One weakness is that it cannot easily be used to compare between different sports, as some sports allow more lopsided results than others. Due to differences in scoring systems, it is easier to double the score of your opponents in sports such as soccer than it is in cricket or basketball.

5.4: Measures for Hypothesis Four (Australia vs New Zealand)

Hypothesis Four compares the five Australian teams against the five New Zealand teams to determine the presence of geographical based superiority. To test this hypothesis, descriptive statistics, lifetime achievement measures, and ladder scores will be used. Furthermore, to account for the possibility of biases arising from an unbalanced schedule, a hypothetical league where every team played one another once was created. This new league can then be analysed using a selection of measures already explained. A further explanation of this method can be found in the results section for Hypothesis Four (B).

5.4.1: Ladder Scores

The ladder score system can be used to determine superiority of groups of teams rather than individually. It is calculated by assigning each team a score according to their rank at the conclusion of each season, i.e. the top team scores a one, the second best team a two, and so on to the bottom score of 10. From this position, a number of options can be explored. By summing the values for each country, it is possible to determine which country performed better. The lower the score, the more superior it is, as it reflects a country's teams finishing higher on the ladder. By dividing the summed value by the maximum, the League Score Average can be found. Finally, The Ladder Score Index is calculated by dividing the most uncompetitive outcome (the one where one country occupies the top five places) by the actual outcome, giving a score between one (the country occupies the top five rankings), and 0.38 (a country occupies the five bottom rankings).

6: Results and Discussion

6.1: Hypothesis One: Concentration

Hypothesis 1: The Suncorp Super Netball league was more competitively balanced in terms of concentration compared to the ANZ Championship.

Concentration refers to the closeness of teams within a league once a season has finished, regardless of their identity. To test this hypothesis five measures were calculated:

- HHI
- RSD
- NAMS
- Concentration Ratios
- The Surprise Index

These measures were chosen for their insight, ability to be adapted for leagues of different sizes, and their reputation within the academic literature.

6.1.1: Herfindahl-Hirschman Index

HHI is a measure of concentration, ranging from one (total imbalance), to zero (perfect balance).

There are a number of iterations of HHI, such as the nHHI, which normalises the results and allows for comparisons between leagues of different sizes. As the Suncorp Super Netball league was two clubs smaller than the ANZ Championship, it is the most valid measure to use.

Results

Table 5 Results of nHHI (Concentration)

	ANZ (2008-2016)	SSN (2017-2018)	Difference
nHHI	0.63	0.65	0.02

Two points are apparent in this calculation. According to nHHI, the ANZ championship is more balanced than the Suncorp Super Netball league. This may come as a surprise, as one of the main justifications for dissolving the old league was to improve competitive balance. The second finding is that the differences between both leagues are very small. Further evidence is required to conclude that the ANZ league is substantially more competitive in regards to concentration in comparison to its successor.

6.1.2: Standard Deviations

Standard deviations measure the dispersion of a dataset relative to its mean, and are calculated as the square of the variance. They are used in calculating two measures of competitive balance, the first being RSD, and the second is NAMS. Both are calculated for each league, and then averaged for each league for comparison purposes.

Relative Standard Deviation

Results

RSD is calculated by dividing the actual standard deviation by the ideal standard deviation. A value of one indicates perfect balance, and the higher the value, the lower the balance.

Table 6: Relative Standard Deviations (Concentration)

	ANZ (2008-2016)	SSN (2017-2018)	Difference
RSD	1.93	2.18	0.25

Table six shows that the ANZ Championship is more balanced in terms of concentration than the Suncorp Super Netball League. This result corroborates with the findings of the HHI, and challenges the original hypothesis. However, it should be noted that RSD is not scalable, and thus comparing leagues of different sizes is not entirely accurate. To accommodate for this characteristic, the NAMS measure can be used, as it normalises the results and allows for comparisons between leagues of different sizes to be more accurately made.

6.1.3: National Measure of Seasonal Imbalance

NAMS is an extension of standard deviations which normalises the results between zero (completely balanced) to one (completely unbalanced). Due to the normalisation process, it is possible to compare leagues of different sizes easily.

Results

Table 7: NAMS Scores (Concentration)

League	ANZ (2008-2016)	SSN (2017-2018)	Difference
NAMS Score	0.78	0.85	0.07

These results reflect those found earlier in the analysis of nHHI; the ANZ Championship was in fact more competitively balanced than its replacement, the Suncorp Super Netball League. This difference is relatively small at -0.07, and thus strong conclusions cannot be easily drawn.

6.1.4: Concentration Ratios

Concentration ratios measure a subset of teams against a relative statistic. By measuring how many of the available points teams within the top k rankings accumulate, it is possible to determine how concentrated a league is.

For this thesis, the concentration ratio was used four times. C1 and C4 are based on Koning's Achievable Attainment ratio, and use the top one and top four teams respectively as the comparative subset. The C5 measure emerges from Michie and Oughton's work on C5 ratio, and expands the subset to the top five clubs. Finally, the NCR is normalised, with upper and lower bounds of one and zero respectively.

Results

Table 8: Results of Concentration Ratios (Concentration)

Average	ANZ (2008-2016)	SSN (2017-2018)	Difference
C1	0.19	0.25	0.06
C4	0.63	0.91	0.28
C5	0.76	0.85	0.09
NCR	0.13	0.75	0.63

In all four cases, the Suncorp Super Netball League is less balanced than the ANZ Championship. Due to its ability to accurately measure leagues of different sizes, NCR is the most useful. In this instance, the difference of 0.63 between the Suncorp Super Netball competition and the ANZ Championship is significant.

These results also suggest that the ANZ Championship had very low levels of concentration. This contradicts Netball Australia's belief that the league was unsustainable due to lack of evenness. While the results from dominance and in-game balance show there is some merit to this argument, it should be noted that in terms of concentration, the ANZ Championship was relatively balanced.

6.1.5: Surprise Index

The Surprise Index measures the amount of upset victories, or “surprises” within a league. A “surprise” is defined as a game where the winner ultimately finished lower than the loser in the final league table. A more thorough explanation of the calculations can be found in the method section.

Results

Table 9: Surprise Index results (Concentration)

Measure	ANZ (2008-2016)	SSN (2017-2018)	Difference
Surprise Index	0.16	0.30	0.14

The Surprise Index finds that the Suncorp Super Netball League had more surprises than the ANZ Championship, and was thus more competitively balanced than the ANZ Championship. This goes against the findings from the previous measures on concentration. The reasons for this are unclear, although the rule change that allows games to finish in a draw, rather than go to extra time, increases the surprise index (all draws are considered surprises, albeit only worth half as much as a surprise win). Another explanation may be the way the Surprise Index is calculated. To grow the index, there either needs to be an increase in the number of games which are classified as a surprise, the distance between the ranked teams needs to widen, or both. Table ten shows the number of surprise games per season, and Figure One illustrates the frequencies of rank differences.

Figure 1: Frequencies of rank differences in surprise index

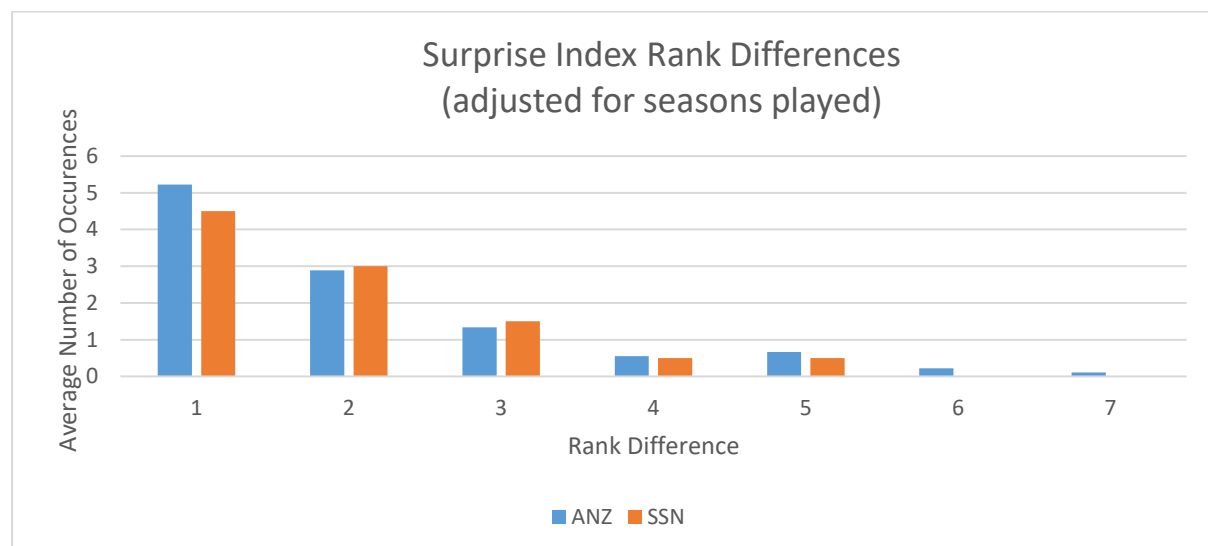


Table 10: Number of games ending in a surprise

Year	Surprises per Year	As % of Games Played
2008	17	26.15%
2009	11	16.92%
2010	12	18.46%
2011	8	12.31%
2012	12	18.46%
2013	11	16.92%
2014	14	21.54%
2015	16	24.62%
2016	14	21.54%
2017	13	25.00%
2018	15	28.85%
Average ANZ	12.78	19.66%
Average SSN	14.00	26.92%
Total	143	20.98%

Figure one shows that differences in ranks are distributed very similarly to the power law, with some minor discrepancies³, and it is hard to separate the two leagues in any meaningful way. Table ten highlights the volatility from year to year; from 2013 to 2014 the number of surprise games doubled. The average of games played that ended in a surprise was higher for the Suncorp Super Netball league (7.26% increase), which may explain the competition's higher Surprise Index. Due to the volatility of the number of surprises per year, the lack of data available for the Suncorp Super Netball league weakens these conclusions.

³ It is anomalous for there to be more surprises with a ranking difference of five (seven occurrences) than a ranking difference of four (six occurrences).

6.1.6: Conclusion

This section examined the hypothesis that the Suncorp Super Netball Championship was more competitively balanced in terms of concentration compared to the ANZ Championship. To test this theory, five measures were used, and their meta results are shown in table 11.

Table 11: Meta Analysis of Hypothesis One (Concentration)

Meta Analysis of Concentration	ANZ Championship	Suncorp Super Netball
Herfindahl-Hirschman Index		
Relative Standard Deviations		
NAMSI		
Concentration Ratios		
Surprise Index		
Total		

Key:



More Unbalanced



More Balanced

As can be seen, the ANZ Championship appears to be the more competitively balanced league in regards to concentration. It is not just the fact that four of the five measures used support this claim, but rather the nature of the measures that are supportive. NAMSI, HHI and the NCR are completely scalable, and allow direct comparisons between leagues of different sizes. The Surprise Index, while it contradicts this statement, is nonetheless weakened by the high volatility regarding the number of surprises from year to year.

The reasons for why the Suncorp Super Netball league is more concentrated are unclear. One explanation may be the poor performance of Adelaide, which has won only a single game in the two years of the competition. Having a single team almost monopolise losing reduces competitive balance. However, the Canterbury Tactix and Central Pulse were perennial strugglers in the ANZ Championship, and this does not appear to have had the same effect. At the other end of the table, the ANZ Championship experienced seasons where teams went undefeated, such as Queensland in 2011. This scenario has yet to occur in the Suncorp Super Netball league.

In summary, the hypothesis, that the Suncorp Super Netball league is more balanced than the ANZ Championship in regards to concentration, is not supported.

6.2: Hypothesis Two: Dominance

The Suncorp Super Netball league was more competitively balanced in terms of dominance compared to the ANZ Championship.

The literature supports the belief that fans are not only interested in close games and close seasons, they are also motivated by the chance to witness their chosen team achieve ultimate success. This phenomenon is analysed using competitive measures of dominance, which unlike concentration measures, takes into account the identity of the team. A league that does not have dominant teams gives fans, players and stakeholders belief that the next season may be “their season”, and are thus more attracted to the game. Leagues where this is not the case, such as Germany’s Bundesliga which has seen Bayern Munich win for the previous eight seasons, have been accused of being dull and stale (Ford, 2018).

A perennial issue with this thesis has been the lack of data, particularly for the Suncorp Super Netball competition. Therefore, some of the measures are not particularly persuasive, although this will change as more seasons are completed in the future. Secondly, both leagues are quite small in terms of number of teams, which makes some measures less reliable. For example, to make the top four in the 20 team English Premier League requires a season of exceptional performance, whereas in a league of eight that barrier is significantly lower.

The three measures used to calculate dominance are the LAM, Adjusted Churn and the HHI.

6.2.1: Lifetime Achievement Measure

LAM is the standard deviations away of a team’s actual win/loss ratio to the idealised ratio of 0.5.

This measure becomes more precise the more seasons that have been completed, which weakens the strength of the Suncorp Super Netball results.

The results for both league are shown in tables 12, 13, and 14.

Table 12: LAM for ANZ Championship

ANZ	Lifetime Achievement
Queensland Firebirds	4.68
Melbourne Vixens	3.74
Waikato Bay of Plenty Magic	3.43
New South Wales Swifts	2.78
Adelaide Thunderbirds	1.60
Southern Steel	-0.19
Northern Mystics	-1.00
West Coast Fever	-2.85
Central Pulse	-5.34
Canterbury Tactix	-7.20

Table 13: LAM for Suncorp Super Netball

SSN	Lifetime Achievement
Sunshine Coast Lightning	2.27
Melbourne Vixens	2.16
Giants Netball	2.01
Queensland Firebirds	0.98
Collingwood Magpies	-0.59
West Coast Fever	-0.76
New South Wales Swifts	-1.41
Adelaide Thunderbirds	-4.54

Table 14: Descriptive Statistics for LAM

	ANZ	SSN	Difference
Min	-7.20	-4.54	-2.66
Max	4.68	2.27	2.41
Range	11.87	6.80	5.07
Std. Dev	3.84	2.19	1.65

Discussion

Three conclusions can be drawn from the results. Firstly, the Suncorp Super Netball league is more competitively balanced than the ANZ Championship. Its worst underachiever, Adelaide, and best overachiever, the Sunshine Coast, are both closer to the mean than their equivalents in the ANZ Championship. The range is smaller (difference = 5.07), as is the standard deviation (difference = 1.65). While Suncorp Super Netball has little data to draw on, the information we do have points towards Suncorp Super Netball being the more competitive league.

Secondly, it is clear that both leagues have either one and two teams which are simply uncompetitive. Adelaide in the Suncorp Super Netball have finished bottom both years, while the Central Pulse and the Canterbury Tactix struggled for most of the ANZ Championship.

In 1992, Quirk and Fort calculated the LAMs of every team in all major US sporting codes between 1901 and 1990. They found the following:

Table 15: Table of top overachievers and underachievers in US Sports

Competitive Balance in Sports Leagues			
Table 7.11 Top Overachievers and Underachievers Ranked by Achievement Measure, All Sports Leagues			
Top Overachievers		Top Underachievers	
Team	Standard Deviations over .500 ^a	Team	Standard Deviations over .500 ^a
1. N.Y. Yankees	15.6	1. Boston Braves	-12.9
2. Boston Celtics	15.4	2. St. Louis Browns	-12.0
3. L.A. Lakers	13.7	3. K.C. Athletics	-8.6
4. Montreal Canadiens	13.1	4. Cleveland (NBA)	-7.3
5. Milwaukee Bucks	9.2	5. Vancouver Canucks	-7.1
6. N.Y. Giants (NFL)	8.5	5. N.J. Nets	-7.1
7. Philadelphia 76ers	7.7	6. Washington Senators I	-6.9
8. Chicago Bears	7.5	7. Washington Senators II	-6.8
9. Boston Bruins	7.1	7. N.Y. Americans (NHL)	-6.8
10. Philadelphia Flyers	6.7	8. San Diego (NFL)	-6.7
11. L.A. Dodgers	6.6	9. Tampa Bay Buccaneers	-6.1
12. Miami Dolphins	6.5	10. Chicago Cardinals	-5.6
13. Edmonton Oilers	6.0	11. New Orleans Saints	-5.0
14. Milwaukee Braves	5.7	12. Pittsburgh Penguins	-4.9
15. Baltimore Orioles	5.6	12. Detroit Pistons	-4.9

Sources: As in table 7.1.

^aUnder equal playing strengths assumption.

4

The Canterbury Tactix, with their score of -7.20, would rank the fifth worst team in American professional sports from 1901 to 1990. A potential reason for this weakness is the earthquake that occurred in 2012, which made it difficult for the club to attract players and fans. This argument is weak, as the local rugby union team, the Canterbury Crusaders, appeared unaffected by the disaster, qualifying for the Super Rugby semi-finals from 2012 to 2014.

⁴ (Quirk and Fort, 1992, p.269)

Finally, it is apparent that there is a significant divide between the New Zealand teams and the Australian teams' ability to win. Apart from the Waikato Bay of Plenty Magic, the top five teams are all Australian. This suggests that Australian teams were better than their New Zealand rivals, a hypothesis which is formally tested later in this thesis.

6.2.2: Adjusted Churn

Adjusted churn is a time series association which divides the actual churn from the previous season by the maximum churn possible. It ranges from a score of zero, which indicates a completely ossified and unbalanced league, to one, which is completely reversed.

Results

Figure 2: Adjusted churn values (Average)

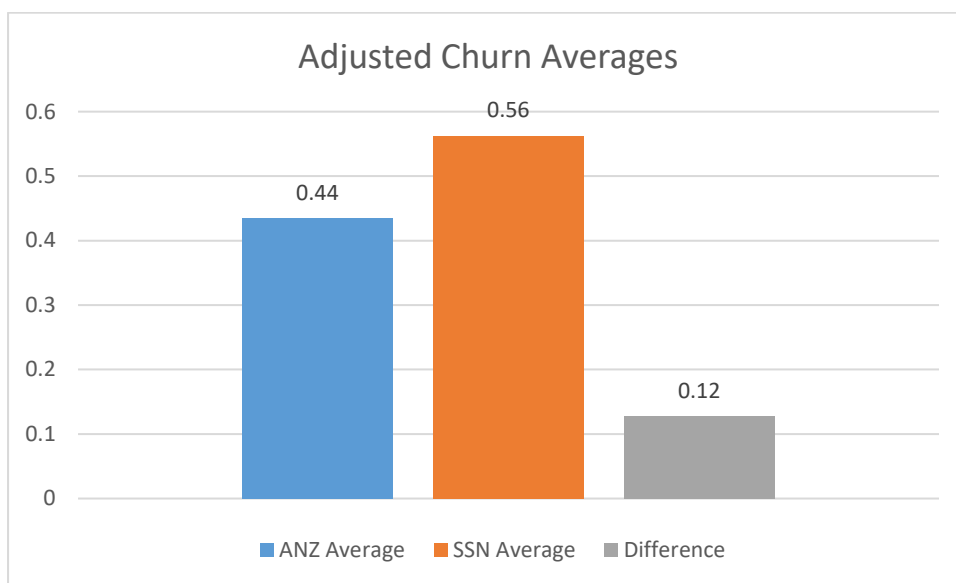
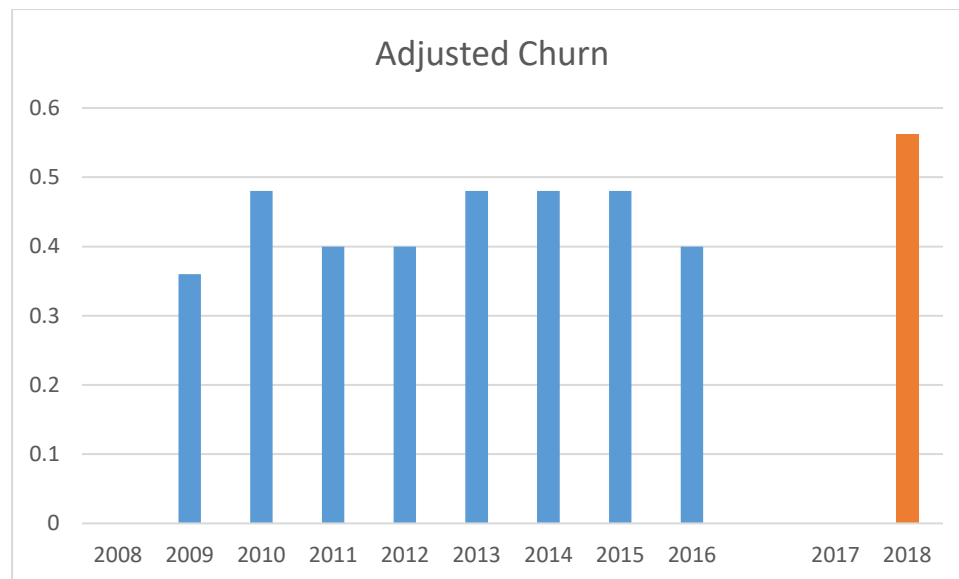


Figure 3: Adjusted Churn Values (Year-by-year)



Discussion

The adjusted churn in 2018 is higher than all years within the ANZ Championship, suggesting the Suncorp Super Netball League has a higher level of competitive balance.

6.2.3: Herfindahl-Hirschman Index

The HHI, previously used to calculate concentration, can be adapted to measure dominance. By summing the squares of a particular criteria, e.g. championships won, top four ranking, finishing last, it is possible to calculate how much teams “monopolise” these criteria. The more data that is available, the more insightful this process becomes, making this the weakest of the three dominance measures used. For example, the Sunshine Coast Lightning have won the premiership in both 2017 and 2018, giving them an HHI score of one, i.e. total monopoly. While winning back to back is an impressive achievement, it is not overly insightful. If Sunshine Coast had won 10 years in a row, that would suggest that they were an incredibly dominant club, and that the league would be advised to intervene to balance out the league. The data, while not particularly illuminating, is displayed in table 16.

Table 16: Herfindahl-Hirschman Results (Dominance)

Summary	ANZ	SSN	Difference
HHI (Top)	0.15	0.19	0.04
HHI (Bottom)	0.15	0.19	0.04
HHI (Premiers)	0.21	0.50	0.29
HHI (Champions)	0.23	1.00	0.77
HHI (Last)	0.38	1.00	0.62

Discussion

The Suncorp Super Netball tournament is simply too small a dataset to draw any significant conclusions from; however, there are insights available in the ANZ Championships. The Sunshine Coast Lightning won back to back championships, which, while nonetheless impressive, is probably less dominant than its HHI score of 1.00 suggests.

As determined from the LAM, the Canterbury Tactix were particularly bad, and thus were frequently at the bottom of the table throughout the ANZ Championship. This is reflected in the HHI scores. While there is low monopolisation of making it into the top or bottom four of the table (1.47 and 1.45 respectively), it is clear that this is a league where a few really bad teams frequently occupy the last position. The HHI (Last) value for the ANZ Championship is 0.383, and is more than double the equivalent for teams dominating first place. This scenario is highly undesirable, as it reduces competitive balance, while repulsing fans of the underperforming team.

6.2.4: Conclusion

Dominance measures focus on the “aspects of competitive balance related to the relative performance of particular teams or groups of teams over a number of seasons rather than on closeness of a league competition” (Evans 2012, p. 33). Another way of describing dominance is that it is the macro level of uncertainty. While it is well and good to have uncertain games, if fans of a particular club know they will never win a premiership, or worse, destined to finish last into the foreseeable future, then fans’ interest is likely to drift. Having a league with low levels of competitive imbalance is in the best interests of all stakeholders.

Table 17: Meta Analysis for Hypothesis Two (Dominance)

Meta Analysis of Dominance	ANZ Championship	Suncorp Super Netball
Lifetime Achievement		
Time Series Associations		
Herfindahl-Hirschman Index	?	?
Total		

While there was very little data available, particularly for the Suncorp Super Netball League, some insights were obtained. For the LAM values, it was clear that the Suncorp Super Netball was more competitively balanced than its ANZ counterpart, primarily due to the weakness of many of the New Zealand teams. The Canterbury Tactix in particular were beneath the required level of

competitiveness in the old league. The “Adjusted Churn Ratio” confirmed the view that the Suncorp Super Netball league was more competitive than the ANZ Championship. The Herfindahl-Hirschman Index was useful in that it allowed deeper insights into the ANZ Championship, but lacked sufficient data to draw conclusions on the Suncorp Super Netball league. For this reason, it is considered inconclusive, although this will change in the future as more seasons are completed. In summary, it can be inferred that the Suncorp Super Netball is the more competitively balanced league in regards to dominance. This assertion is mitigated by the lack of data available, and further study of this area should be conducted in years to come when more data is available.

6.3: Hypothesis Three: In-Game Evenness

Games in the Suncorp Super Netball league were more competitive than games in the ANZ Championship.

The uncertainty of outcome theory posits that fans are attracted to the unpredictable nature of sporting contests, and this is widely supported by the academic literature. Perhaps more importantly, sporting competitions act in accordance with this belief. Mechanisms such as reverse-order drafts, salary caps and limitations on labour movement are all symptoms of organising bodies aiming to increase match uncertainty. It is therefore logical to assume that more competitive games are more uncertain, and thus more attractive to the fans and associated stakeholders. Given the paucity of completed season, these measures can be particularly insightful. To measure within-game competitiveness, three measures are used. They are:

- Descriptive statistics
- The “Blow-Out index”
- The ARIPD measure.

A brief conclusion summarises the findings and addresses whether the hypothesis is supported.

6.3.1: Descriptive Statistics

The two major descriptive statistics are the mean and the median. The lower these values are, the closer the games and more competitive the competition.

Table 18: Descriptive Statistics (In-Game Balance)

Margins	ANZ	SSN	Difference
Average Margin	9.69	8.84	-0.85
Median Margins	8	7	-1

The results clearly show that the ANZ Championship games had larger margins by an average of 0.85 points. This suggests that the Suncorp Super Netball games were marginally more competitive, although that margin is very small.

The median margin is smaller in the Suncorp Super Netball (7) than in the ANZ Championship (8). Again, a smaller value here suggests more games finish with close margins, a hallmark of a competitive competition.

6.3.2: Percentages

As explained in the method section, percentages are calculated as

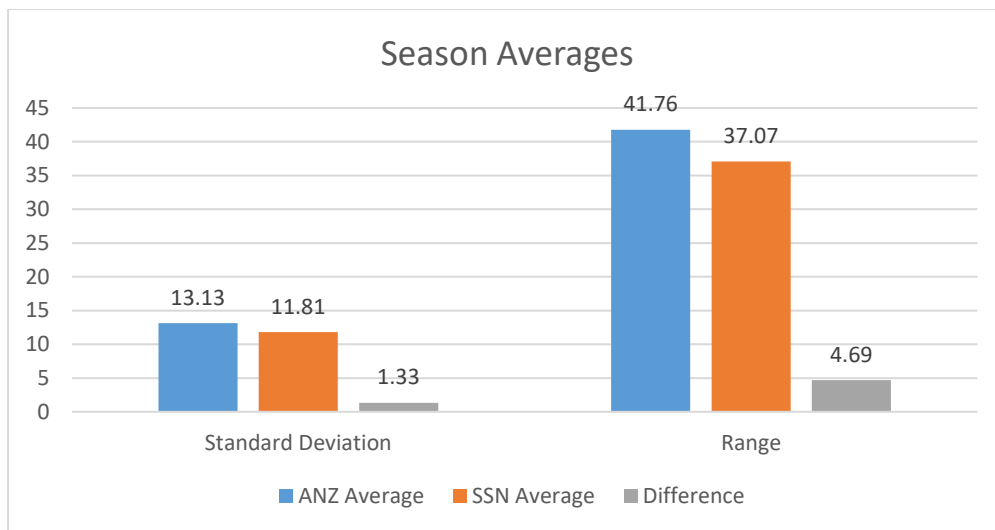
$$\frac{\text{Sum of goals scored}}{\text{Sum of goals conceded}} \times 100$$

Percentages can be used as a measure of competitive balance within a season in two different ways. The first is by using descriptive statistics, the second is by comparing the actual league results with an ideal league. This section explains this in further depth as well as providing a discussion to the results calculated.

6.3.3: Descriptive statistics

The two major descriptive statistics derived from the percentage are the standard deviations and the range. In both instances, the averages of each seasons are combined to find a league mean, which is what will be used for comparison purposes.

Figure 4: Descriptive percentages regarding statistics



Range

Range is the difference between the maximum and minimum points. To calculate this, the mean of all seasons was found, which was then averaged to find the league lifetime mean, with the results shown in table 19.

Table 19: Range Results (In-Game Balance)

	ANZ Average	SSN Average	Difference
Range	41.76	37.07	-4.69

The range in the Suncorp Super Netball was on average 4.69 points smaller than the ANZ average. The smaller the range, the closer the teams are in terms of ability, and the greater the level of competitive balance. In this instance, we can conclude that the Suncorp Super Netball league is more balanced.

6.3.4: Standard Deviation

Standard deviations measures the level of dispersion relative to its mean, which in this example is 100%. The smaller the standard deviation, the closer the teams are to the “ideal” league, and thus the more competitively balanced the league is. By following the steps outlined in the range, it was possible to calculate the standard deviations for the lifetimes of the league, and compare them against each other. In this instance, the following values were found.

Table 20: Standard Deviations (In-Game Balance)

	ANZ Average	SSN Average	Difference
Standard Deviation	13.13	11.81	-1.33

Once again, the Suncorp Super Netball League has a lower standard deviation by 1.33, and thus is more competitively balanced. As stated previously, the lower the standard deviation, the closer the league's teams are to the ideal league, which is the league that has maximised talent equality.

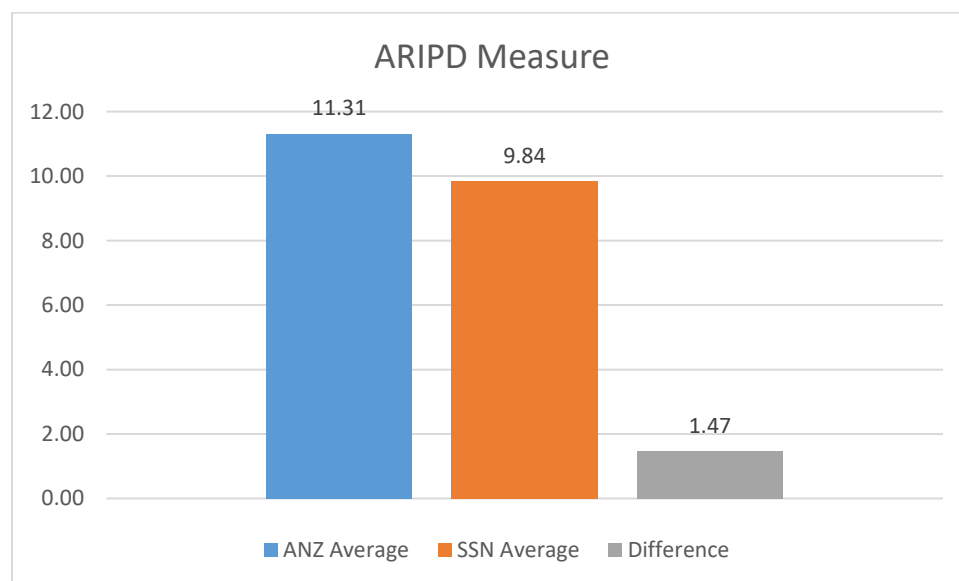
When using both range and standard deviation to measure competitive balance, on both occasions it was found that the Suncorp Super Netball League was more competitively balanced.

6.3.5: Average of Real to Idealised Percentage Differences

The third measure of using percentages as an indicator of competitive balance is the ARIPD measure.

When measuring the ARIPD scores for both the ANZ and Suncorp Super Netball leagues, the results are as follows:

Figure 5: Comparison of ARIPD values



The Suncorp Super Netball had a lower ARIPD score of 9.84, compared to ANZ which had a value of 11.31. This supports the previous conclusions that when viewed through the prism of “in-game competitive balance”, the Suncorp Super Netball league is more balanced.

6.3.6: Blow-Out Index

The “Blow-Out index” is the percentage of games in a season where the final margin is in excess of ten points. The results were calculated for each season and then averaged to find the league average. Below are the findings:

Figure 6: Blow-Out Index Results

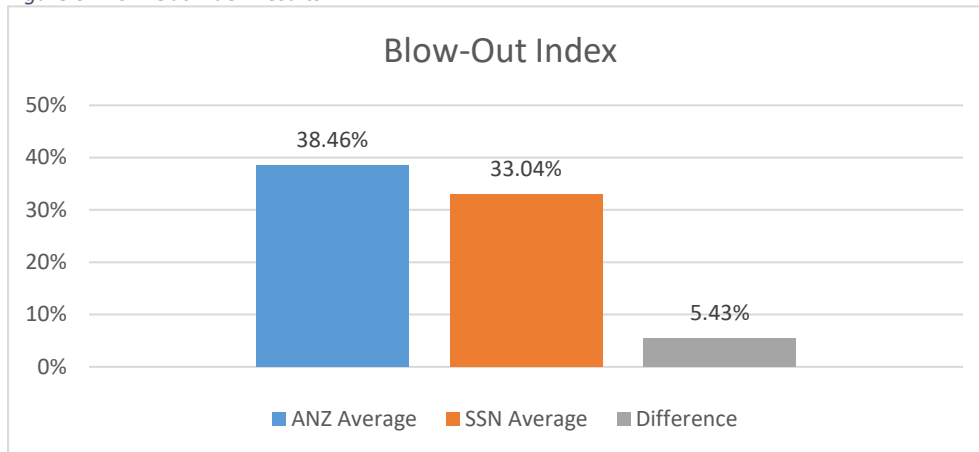


Table 21: Blow-Out Index Results

Blow-Out Index	ANZ Average	SSN Average	Difference
Average Blow-Out (%)	38.46%	33.04%	-5.43%

The results show that there are more Blow-Outs in the ANZ championship than the Suncorp Super Netball league by 5.43%. This suggests that the Suncorp Super Netball is the more competitive league.

6.3.7: Conclusion

Table 22: Meta Analysis of Hypothesis Three (In-Game Balance)

Meta Analysis of In-Game Balance	ANZ Championship	Suncorp Super Netball
Descriptive Statistics		
Blow-Out Index		
ARIPD		
Total		

In all measures tested, the ANZ Championship was less competitively balanced than the Suncorp Super Netball League in regards to in-game balance.

6.4: Hypothesis Four (A): Australia v New Zealand

The Australian teams are stronger⁵ than the New Zealand teams in the ANZ Championship.

6.4.1: Introduction

When the ANZ Championship disbanded after the 2016 season, one of the primary reasons was the belief that the New Zealand teams were uncompetitive. While the Waikato Bay of Plenty Magic was a perennial finals contender, and were crowned champions in 2012, the Central Pulse and Canterbury Tactix almost always finished last. To test the hypothesis that the Australian teams were better than their New Zealand counterparts, three measures are used. Firstly, descriptive statistics of margin, championships and games won head to head are analysed. Secondly, league rankings are scored and averaged to determine which country has enjoyed more successful seasons. The final measure is the average LAM for each country. A short summary is provided at the end.

6.4.2: Descriptive Statistics

The three descriptive statistics used are the number of championships won, the number of games won head to head, and the average margin for each country when they did win. The results are provided below in tables 23, 24 and 25.

Table 23: Number of Championships won

Country	Number of Championships Won
Australia	8 (2008-2011, 2013-2016)
New Zealand	1 (2012)

Table 24: Number of games won head to head

Country	Number of Games won Head to Head
Australia	165
New Zealand	60
Draws	4

Table 25: Average winning margin (Head to Head)

Country	Average winning margin Head to Head
---------	-------------------------------------

⁵ For the purposes of this thesis, the terms superior, better, and stronger are considered interchangeable.

Australia	12.85
New Zealand	6.61

Table 25 measures the average winning margin for the when Australia beat New Zealand, and vice-versa. It does not include domestic games i.e. Melbourne Vixens vs Adelaide Thunderbirds.

All three tables outline a strong suggestion that the superior teams came from Australia. They won more championships, more games, and they won those games by greater average margins. While descriptive statistics are by nature blunt instruments, and thus do not tell the whole story, they do infer very strongly in this case that the Australian teams were significantly stronger than their New Zealand counterparts in the ANZ Championship.

6.4.3: Ladder Score Averages

By assigning clubs points based on their end of year rankings and then summing them based on their country of origin, it is possible to compare the various levels of success between both countries. The team that finishes first is assigned a score of one, the second best team scores two and so on, thus the lower the scores the better. The ladder score percentage is calculating by dividing the sum of scores by the total number, and again a lower score indicates a stronger subset. The ladder score is explained more comprehensively in the method section, and unlike the two preceding measures, confers that a higher score indicates a better performance.

Table 26: Ladder Score Averages

	Sum of Scores	Ladder Score Percentages	Ladder Score Index
Australia	23	42%	66%
New Zealand	32	58%	47%

This measure supports the notion that Australian teams were better than their New Zealand counterparts.

6.4.4: Lifetime Achievement Measure Averages

By averaging the LAMs for each country, it is possible to determine whether the Australian teams were weaker than the New Zealand teams. The full list of LAMs is found on table 12.

Results

Table 27: LAM Averages for Australia v New Zealand

	Australia	New Zealand	Difference
LAM Averages	1.72	-2.13	-3.85

Discussion

The LAM averages suggest that Australian teams were stronger than their New Zealand rivals.

6.4.5: Conclusion

Meta Analysis of Australia v New Zealand	Australia	New Zealand
Descriptive Statistics	★	✗
League Score Average	★	✗
LAM Averages	★	✗
Total	★	✗

6.5: Hypothesis Four (B): Even Schedule

6.5.1: Introduction

In many sports leagues, particularly in European soccer, every team plays every other team at home and away, and the team at the top of the table at the season's end is the undisputed champion. However, the ANZ Championship used an uneven playing schedule, which reduced the inherent "evenness" of the competition. Teams played against their domestic opposition twice, but only competed once against the international clubs per season. For example, the Central Pulse, a New Zealand team, played against the Canterbury Tactix at home and away, but only once against the Melbourne Vixens. The reasons for doing this were understandable; international travel costs placed a significant financial burden onto the clubs, and domestic games were more popular than trans-Tasman contests (Mitchell, 2016). However, whenever a competition doesn't have a fair schedule, there will be teams that benefit from playing weaker teams proportionally more than their opposition, thus threatening the integrity of the competition.

It is possible that the unbalanced schedule biased the teams based on their geography. By creating a hypothetical ANZ Championship league where every team plays other an equal number of times, this bias would be removed, allowing clearer insights to be made. This section is made up of three parts, with the first explaining how this hypothetical league was calculated. The second section calculates and contrasts the new league with the original league by using a sample of the measures previously used in this thesis. The final section is a brief discussion of the results and the limitations of this experiment. A selection of the data is found in Appendix Three (Section 11.3).

6.5.2: Method

Table 28: Explanation of Hypothetical League Changes

Game	Home Team	Score (Home Team)	Away Team	Score (Away Team)
Game 1	Team A	20	Team B	22
Game 2	Team B	8	Team A	20
“Virtual” Game	Team A	$20 = [(20+20)/2]$	Team B	$15 = [(22+8)/2]$

In this imaginary league, Team A and Team B play each other home and away. While both teams win a game apiece, when the average of each score is taken, Team A (Score of 20) is the winner over Team B (Score of 20). This process was completed for each season of the ANZ, which resulted in a new, virtual league where every team played 9 times, for a total season of 45 games. These results were then used to construct virtual league tables, which can be compared against the original leagues for measures of concentration, dominance, and in-game balance. Due to the limited scope of this hypothesis, only a subset of available measures were used. These were chosen for their insightfulness and ease of comparability, and are as follows:

- NAMS (Concentration)
- HHI(Concentration)
- LAM (Dominance)
- Adjusted Churn (Dominance)
- Blow-Out index (In-game balance)

6.5.3: Results

National Measure of Seasonal Imbalance

Table 29: NAMS Results for Hypothetical League Two

	2008	2009	2010	2011	2012	2013	2014	2015	2016	Average
ANZ (Even Schedule)	0.79	0.84	0.87	0.95	0.72	0.84	0.67	0.90	0.96	0.84
ANZ (Original)	0.74	0.88	0.85	0.87	0.74	0.80	0.59	0.73	0.87	0.78
Difference	-0.06	0.04	-0.02	-0.08	0.02	-0.04	-0.07	-0.17	-0.09	-0.05

Figure 7: Comparison of NAMSIS Measure between

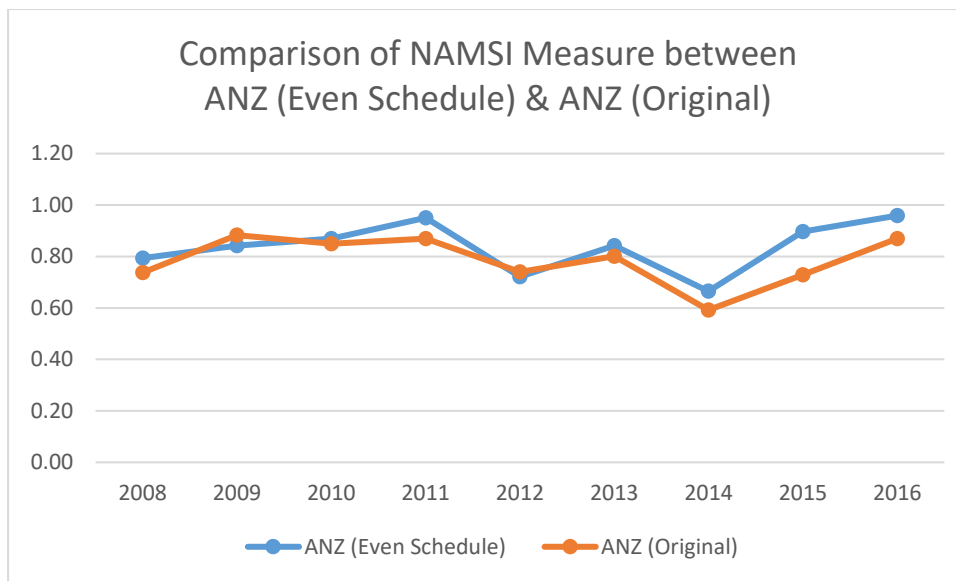
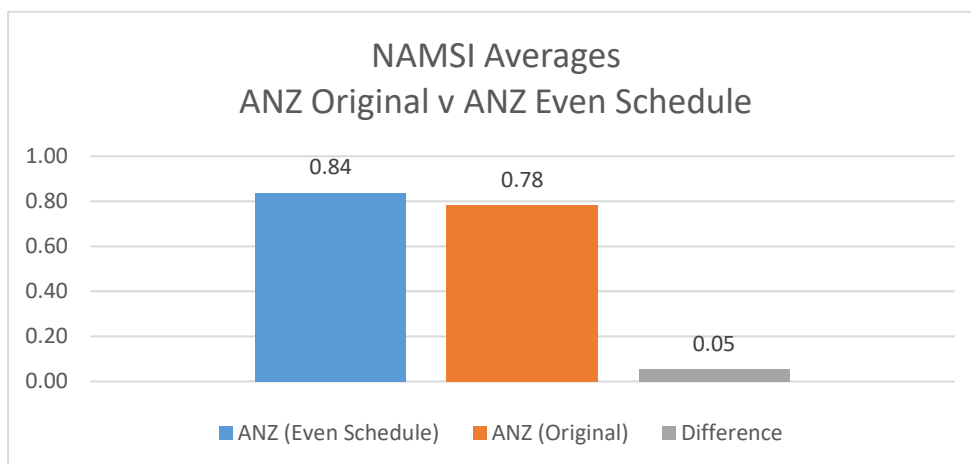


Figure 8: NAMSIS Averages (ANZ Original v ANZ Even Schedule)



Aside from the seasons in 2009 and 2012, the original ANZ league had a lower NAMSIS score than the new virtual leagues. NAMSIS is a normalised measure bounded by zero (complete balance) and one (complete imbalance). The seven of the nine seasons played, as well as the average, show the ANZ (Even Schedule) competition being more imbalanced than the original.

Herfindahl-Hirschman Index (Concentration)

Table 30: NAMSJ Results for Even Schedule League

Averages	ANZ (Even Schedule)	ANZ (Original)	Difference
HHI	0.128	0.126	-0.002
dHHI	0.028	0.026	-0.002
nHHI	0.202	0.182	-0.020

In all three iterations of the HHI, the new hypothetical league returns higher values. This suggests that it is more imbalanced in terms of concentration. Note that in table 30 values are calculated to three decimal points. This is account for the small differences between leagues.

Blow-Out Index

Table 31: Descriptive Statistics for Even Schedule League

Score	ANZ ES	ANZ Original	Difference
Number of Games	405	592	187
Number of Blowouts	165	228	63
Blow-Out Index	40.70%	38.51%	-2.19%

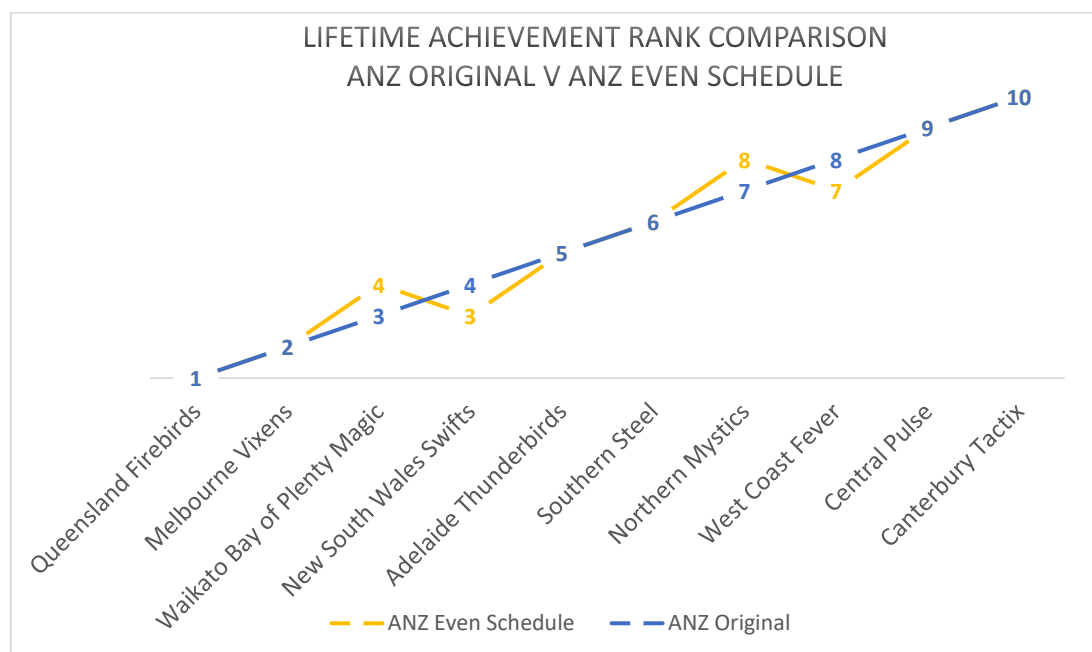
In the new hypothetical league, there is a 2.19% increase in games that end in a Blow-Out. As previously determined, Blow-Outs indicate games with very low outcome uncertainty, and thus are of little interest to fans. The reason for this increase emerges from hypothesis four, which states that the Australian teams were superior on average to their New Zealand counterparts. In the original ANZ league, there were more domestic matches, and therefore it would be expected that there were more games featuring teams of similar standards. This would naturally reduce the number of Blow-Outs as a proportion of games played.

Lifetime Achievement

Table 32: Lifetime achievement measures of ANZ to ANZ (Even Schedule)

Teams	ANZ Original	Even Schedule	Difference
Adelaide Thunderbirds	1.60	1.00	-0.60
Canterbury Tactix	-7.20	-6.67	0.53
Central Pulse	-5.32	-4.89	0.43
Melbourne Vixens	3.74	3.89	0.15
New South Wales Swifts	2.78	3.33	0.55
Northern Mystics	-1.00	-2.11	-1.11
Queensland Firebirds	4.68	4.67	-0.01
Southern Steel	-0.19	-0.11	0.08
Waikato Bay of Plenty Magic	3.43	2.11	-1.32
West Coast Fever	-2.85	-1.22	1.63
Australian Average	2.33	1.99	-0.34
New Zealand Average	-2.33	-2.05	0.28
Range	11.87	11.33	-0.54
Standard Deviation	3.84	3.57	-0.26

Figure 9: Lifetime Achievement rank comparisons between ANZ (Original) and ANZ (Even Schedule)



The results calculated provide three insights into the ramifications of having an even league.

1. There is little difference in terms of lifetime achievement at the top end. The Queensland Firebirds are 0.01 points better off in the ANZ Original league compared to the virtual league, suggesting that moving to an even roster would have had very little effect on the club. This trend was not repeated at the bottom of the table, where the Canterbury Tactix were 0.53 standard deviations better off in the hypothetical league. This is driven by the fact that in both leagues, Canterbury would only achieve 21 victories, but the increase in games played in the ANZ Championship lowered their winning percentage.
2. The standard deviation and range both suggest that the ANZ Even Schedule league is more competitive. This is primarily driven by the improvement of the Canterbury Tactix and the Central Pulse in the new league, due to their improved winning percentage caused by playing less games.
3. Finally, the change in ranking from an unbalanced to balanced schedule reveals that the Australian teams were unfairly biased due to the roster. Figure Nine shows the rankings of teams across both leagues from first place (Queensland Firebirds) to last (Canterbury Tactix). It can be seen that two Australian teams, the NSW Swifts and West Coast Fever, both improve a place in the even schedule, from 4th to 3rd and 8th to 7th respectively. At the same time two New Zealand teams, the Waikato Bay of Plenty Magic and the Northern Mystics, regress a ranking place. With a match schedule that was not biased, it becomes apparent that the roster was artificially improving the Win/Loss record of the NZ teams, and punishing the Australian teams.

Time series Associations

Table 33: Time Series Associations for Even Schedule League

Time Series	ANZ Even Schedule	ANZ Original	Difference
Adjusted Churn	0.41	0.44	0.03
Top 4 Reoccurrence	71.92%	62.51%	-9.41%
Correlation Coefficient	56.21%	48.94%	-7.27%

The results for the calculated time series both confirm that the original ANZ league is more balanced than the new virtual league. In regards to adjusted churn, the difference is 0.03, which translates to a percentage change of 7.41%. Similarly, the top four reoccurrence results show that there was increased ossification of teams within the new virtual league. These changes can be explained by the general difference in ability between teams from Australia and New Zealand. By playing more games

against teams from their own country, more matches were even, and this translated to increased competitive balance in terms of dominance.

Conclusion

Five measures were used to calculate the competitive balance of a virtual league which ran on an even schedule. Having an even schedule is used prolifically in European soccer leagues, and removes the possibility of scheduling bias. The results of the five measures are presented in a meta review in table 34.

Table 34: Meta Analysis of ANZ v ANZ (Even Schedule)

Meta Analysis of ANZ v ANZ (Even Schedule)	ANZ Championship (Original)	ANZ Championship (Even Schedule)
NAMSI		
HHI		
Lifetime Achievement		
Time Series Associations		
Blow-Out Index		
Total		

While five measures is by no means extensive, they provide insights into the balance with respect to concentration, dominance and in-game. The results indicate that the original ANZ Championship is more balanced than the new league, with the one exception being the LAM. These results are largely predictable, as they follow on from the previous measures tested in Hypothesis Four A.

The fact that the original league was more balanced than the virtual league further supports the findings found in Hypothesis Four A, which stated that the New Zealand teams were on average weaker than the Australian teams. By playing more games against teams of a more equal strength, the Australian teams were disadvantaged, while the New Zealand team enjoyed the opposite.

If the ambition of the league organisers is to optimise the level of competitive balance, the original ANZ league is a better choice. However, if the ideal is to create the “fairest” schedule, then the ANZ Even Schedule league is superior. When the lifetime table for both leagues are compared, it becomes evidently clear that two New Zealand teams, the Waikato Bay of Plenty Magic and the Northern Mystics, both benefited at the expense of West Coast Fever, and the NSW Swifts. However, this advantage wasn’t particularly strong, as can be seen by the results found regarding Hypothesis Four A.

6.5.4: Conclusion

This hypothesis explored whether the Australian teams were superior to their New Zealand opponents in the ANZ Championship between 2008 to 2016. Were it to be found that the New Zealand teams were overall less talented than their Australian opposition, Netball Australia's decision to break away and form the Suncorp Super Netball league would be more justified. To test this hypothesis, an analysis of descriptive statistics and ladder rankings was conducted. Furthermore, a hypothetical, even scheduled league was created as an experiment to test biases in rostering. In all three, the results strongly support the claim that the Australian teams were superior to their New Zealand contemporaries.

7: Results and Discussion Summary

This section tested the following four hypotheses:

Hypothesis One: The Suncorp Super Netball league is less concentrated than the ANZ Championship

Hypothesis Two: The Suncorp Super Netball league exhibits less dominance than the ANZ Championship

Hypothesis Three: The Suncorp Super Netball league has games which are closer and more competitive than the ANZ Championship

Hypothesis Four (A): The New Zealand teams were weaker than the Australian teams when measured head to head (ANZ Championship only).

Hypothesis Four (B): The uneven playing schedule did not affect the differences in strength between Australian and New Zealand teams.

7.1: Hypothesis One: Concentration

The meta results for this hypothesis are outlined in table 35 as to which leagues are more competitively balanced in terms of concentration.

Table 35: Meta Analysis of Hypothesis One (Concentration)

Meta Analysis of Concentration	ANZ Championship	Suncorp Super Netball
Herfindahl-Hirschman Index		
Relative Standard Deviations		
NAMSI		
Concentration Ratios		
Surprise Index		
Total		

These results suggest that the ANZ Championship was more competitively balanced than its successor, the Suncorp Super Netball League. Four of the five measures support this claim, with only the surprise index suggesting otherwise. As discussed previously, this measure has a high volatility in the number of games that are deemed surprises, which impacts on the veracity of the results found for the Suncorp Super Netball League. As more seasons are played, the results of this index will become more conclusive.

It is perhaps surprising that the Suncorp Super Netball League was found to be less competitive than the ANZ Championship, as one of the purposes of the new league formation was to create a more balanced competition. Reasons as to why this may be the case may be the very poor performance of Adelaide (one win in two years), as well as the even scheduling of the roster.

7.2: Hypothesis Two: Dominance

The meta results for this hypothesis are outlined in regards to dominance in table 36, competitively balanced in terms of concentration.

Table 36: Meta Analysis of Hypothesis Two (Dominance)

Meta Analysis of Dominance	ANZ Championship	Suncorp Super Netball
Lifetime Achievement		
Time Series Associations		
Herfindahl-Hirschman Index	?	?
Total		

All three measures support the claim that in terms of dominance, the Suncorp Super Netball team was more balanced than the ANZ Championship, which is in line with the stated hypothesis. Of all the hypotheses, this one is the weakest, as measures of dominance are strengthened by an increased number of seasons to analyse. As only two editions of the Suncorp Super Netball league have been completed, these results should be treated with caution.

Besides the lack of data, there are some reasons as to why the ANZ Championship featured higher dominance. Canterbury Tactix and Central Pulse were perennial under-achievers, with the former never playing finals. At the other end of the table, the three Eastern Seaboard teams, NSW, Melbourne, and Queensland, as well as the reliable Waikato Bay of Plenty Magic, made up 14 of the possible 18 Grand Final places over the length of the competition.

While the data suggests that the Suncorp Super Netball competition is more balanced in terms of dominance, it should continue to monitor the situation. The Sunshine Coast Lightning have been crowned champions both years, while the Adelaide Thunderbirds have simultaneously finished last. Were this scenario to repeat itself for another three years, Netball Australia would be wise to consider adopting suitable equalisation measures such as a reverse order draft.

7.3: Hypothesis Three: In-Game Balance

The meta-results for this hypothesis in regards to in-game balance are presented in Table 37

Table 37: Meta Analysis of Hypothesis Three (In-Game Balance)

Meta Analysis of In-Game Balance	ANZ Championship	Suncorp Super Netball
Descriptive Statistics		
Blow-Out Index		
ARIPD		
Total		

All three measures support the claim that in terms of in-game evenness, the Suncorp Super Netball team was more balanced than the ANZ Championship. There is a significant amount of cross-over within this hypothesis; a league with more Blow-Outs is more likely to have a larger average margin, and vice-versa.

7.4: Hypothesis Four (A): Australia v New Zealand

Table 38: Meta Analysis of Hypothesis Four (A) (Australian v New Zealand)

Meta Analysis of Australia v New Zealand	Australia	New Zealand
Descriptive Statistics		
League Score Average		
LAM Averages		
Total		

Discussion

All three measures support the claim that the Australian teams were significantly better than their New Zealand counterparties. A potential reason for this may be due to population size. Australia's population (25.13 Million) dwarfs New Zealand's (4.24 million), and this gap was further exacerbated at the city level. For example, Sydney, Australia's largest city, is around 95 times that of the Southern Steel's Invercargill. This difference meant that the New Zealand team had a smaller pool of talent to recruit from, as well as a smaller fan base to collect revenue from. To determine whether this was

the case, a correlation coefficient was calculated between LAM (a measure of dominance) and population size. The value of R is 0.52. This is a moderate positive correlation, which means there is a tendency for high population variable scores to go with high win variable scores (and vice versa).

Table 39: List of teams, locations, populations, and LAM

Club	City	Population ('000)	LAM
Adelaide Thunderbirds	Adelaide	1,313	1.60
Canterbury Tactix	Christchurch	404	-7.20
Central Pulse	Wellington	418	-5.32
Melbourne Vixens	Melbourne	4,850	3.74
New South Wales Swifts	Sydney	5,131	2.78
Northern Mystics	Auckland	1,628	-1.00
Queensland Firebirds	Brisbane	2,408	4.68
Southern Steel	Invercargill	51	-0.19
Waikato Bay of Plenty Magic	Hamilton	241	3.43
West Coast Fever	Perth	2,050	-2.85

7.5: Hypothesis Four (B): Even Schedule

Table 40: Meta Analysis of ANZ v ANZ (Even Schedule)

Meta Analysis of ANZ v ANZ (Even Schedule)	ANZ Championship (Original)	ANZ Championship (Even Schedule)
NAMSI		
HHI		
Lifetime Achievement		
Time Series Associations		
Blow-Out Index		
Total		

Discussion

To test whether the uneven nature of the ANZ Championship schedule may have influenced the difference in strengths of teams from Australia compared to the New Zealand teams. This was tested by recreating the ANZ Championship that had every team playing each other once, then calculating a sample of competitive balance measures.

The NAMSI, HHI, Time Series Associations and Blow-Out Index all found that the original ANZ Championship was more balanced, while the LAM contradicted this claim. These results suggested that the uneven roster did not create the imbalances between Australian and New Zealand teams,

on the contrary, they minimized the differences. As the New Zealand teams played more games amongst themselves, they were able to artificially inflate their success.

8: Using Hypothetical Leagues to test policy

Testing the first three hypotheses indicated that the Suncorp Super Netball league was more competitive in measures of in-game balance and dominance, while the ANZ league was more competitive in regards to concentration. The final hypothesis offered strong support that the New Zealand teams were weak compared to their Australian opponents, and thus splitting the ANZ Championship into two separate domestic leagues appears sensible. However, this approach may be too simplistic. While it is fair to say teams such as Canterbury Tactix and to a lesser extent Central Pulse were too weak to be in the league, the Waikato Bay of Plenty Magic were one of the top teams in the competition. They qualified for the finals seven times, won the championship in 2012, and were ranked third on the LAM ranking list. By these measures, Waikato Bay of Plenty Magic was capable of competing against Australian teams.

The Suncorp Super Netball has not been as competitively balanced as was hoped. Not only has concentration imbalance increased, but the Adelaide Thunderbirds have struggled severely, losing all games bar one in the past two seasons. There is room to conject that the current format is not the optimal outcome after the ANZ Championship dissolution.

To test potential policy changes, two hypothetical leagues have been created. The first reimagines the ANZ Championship without the Central Pulse and the Canterbury Tactix, two New Zealand teams which were ranked last in the LAM Rankings. The second league the Suncorp Super Netball is replicated with the Adelaide Thunderbirds excluded from the competition. After testing both hypothetical situations, it may be possible to determine whether the decisions that Netball Australia made were optimal, or whether there were better options available.

8.1: Limitations to research

There are some limitations to these hypothetical leagues, as follows:

- It makes the rather large assumption that teams in the new hypothetical league would have identical strengths and results to the real league. This is unlikely.
- It would be assumed that at least some of the players at the disbanded clubs would transfer to different clubs, thus changing the ability of those teams.
- Due to the unbalanced draw in the ANZ Championship, hypothetical league one would require the New Zealand teams to play nine games, while the Australian teams would play 11 games a season. This outcome would be highly unlikely to occur in real life.

- The assumption that removing teams has no effect on the results of other games is weak. Shorter seasons, less travel, and less “easy games” are a small example of the many factors that affect the remaining teams.

8.2: Hypothetical League One: Removing Central Pulse and Canterbury Tactix

The LAM has shown that the Central Pulse and Canterbury Tactix were simply too weak in the old league. This had the effect of lowering in-game competitiveness and increasing dominance and concentration, thus lowering overall competitive balance. In hypothetical league one, the two teams are removed, bringing the league down to eight teams. By deleting all games that included one or both of these teams, it was possible to calculate a selection of measures to determine the change in competitive balance. The measures chosen were:

- Descriptive Statistics (In-Game Balance)
- NAMI (Concentration)
- HHI (Concentration)
- LAM (Dominance)

Table 41: In-game balance results for Hypothetical league one

	ANZ	Hypothetical League One	Difference
Games Played	585	369	-216
Average Margin	9.72	9.06	-0.66
Median Margin	8	7	-1
Number of Blowouts	225	133	-92
Blowout Index	39.56%	36.04%	-3.52%
Average Score	52.33	52.07	-0.26

This table shows that in the new league, the average margin and the Blow-Out index both decreased, which are key factors for in-game competitiveness. It is unsurprising that this was the case, as the two excluded teams, Canterbury Tactix and Central Pulse, were both frequently involved in blow out games. In the ANZ Championship, 39.04% of games that ended in blowouts had either Canterbury or Central as the losing team.

8.2.1: National Measure of Seasonal Imbalance (NAMSI)

NAMSI is a normalised measure of concentration ranging from zero (complete balance) to one (complete imbalance). The results are shown in table 42.

Table 42: NAMSI Results for Hypothetical League One

	ANZ	Hypothetical League One	Difference
NAMSI Average	0.79	0.78	-0.01

According to NAMSI, the change in concentration balance in the new league is minimal.

8.2.2: Herfindahl-Hirschman Index (HHI)

There are a number of different variations of HHI that can be used to measure concentration. As the number of teams change from the ANZ (ten) to the hypothetical league (eight), the best measure to use is the nHHI, which normalises the values between zero (complete balance) and one (complete imbalance).

Table 43: HHI Results for Hypothetical League One

	ANZ	Hypothetical League One	Difference
HHI	0.126	0.158	0.032
dHHI	0.026	0.033	0.007
nHHI	0.182	0.183	0.001

The table shows that the two leagues are very similar in regards to concentration. This is in line with the NAMSI measure, which found minimal differences.

8.2.3: Lifetime Achievement Measure (LAM)

LAM uses standard deviations to measure how much an individual club over or underachieves. A value of zero is equal to a team with a lifetime win record of 50%, positive values indicate over performance, while negative value indicate underperformance. Measurement of the range and standard deviation of the LAM table are used to calculate dominance, in both instances, a lower value equals greater competitive balance.

Table 44: Comparison of Lifetime Achievement for ANZ Championship to Hypothetical League One

	ANZ	Hypothetical League One	Difference
Adelaide Thunderbirds	1.60	-0.30	-1.91
Canterbury Tactix	-7.20	N/A	N/A
Central Pulse	-5.32	N/A	N/A
Melbourne Vixens	3.74	2.11	-1.63
New South Wales Swifts	2.78	2.11	-0.67
Northern Mystics	-1.00	-3.67	-2.66
Queensland Firebirds	4.68	3.12	-1.56
Southern Steel	-0.19	-2.11	-1.92
Waikato Bay of Plenty Magic	3.43	1.44	-1.99
West Coast Fever	-2.85	-4.32	-1.47
Min	-7.20	-4.32	-2.87
Max	4.68	3.12	1.56
Range	11.87	7.44	4.44
St.Dev	3.84	2.67	1.17
NZ Average	-2.05	-1.44	-0.61
Aus Average	1.99	0.54	1.45

In the new hypothetical league, there is an improvement in competitive balance in terms of dominance. The minimum and maximum values are both closer to zero, the ideal value, which in turn reduces the range, a measure of evenness. This is partially due to the removal of the Canterbury Tactix, which were the worst team in the ANZ Championship, and had the lowest LAM value (-7.20). The standard deviation in the new hypothetical league was lower, indicating that the competition had become more balanced. Finally, the dominance of the Australian teams was reduced. Their average of 0.54 is 1.45 standard deviations away from the Australian team average in the ANZ Championship. The New Zealand teams also improved, although by a lesser amount.

8.2.4: Conclusion

Table 45: Meta Analysis of ANZ Championship to Hypothetical League One

Meta Analysis of Hypothetical League One	ANZ Championship	Hypothetical League One
Descriptive Statistics		
NAMSI		
HHI		
Lifetime Achievement		
Total		

In the new hypothetical league, the two lowest performing teams, the Canterbury Tactix and the Central Pulse were removed. By recreating the ANZ Championship sans these two teams, it was possible to test for changes in in-game balance, concentration, and dominance.

The new hypothetical league was more balanced in regards to dominance, and in-game measures, although there was no meaningful change in concentration. The reasons for this are relatively straight forward; the Canterbury Tactix and to a lesser extent the Central Pulse had a significant effect on unbalancing the league. As noted earlier, 39.4% of all games resulting in a blowout in the ANZ Championship featured one of the two teams. Furthermore, the Canterbury Tactix, with their LAM value of -7.20, are a historically weak team. Quirk and Fort made a table of the best and worst performing teams across the four major sports in North America, and Canterbury's results would have placed them as the fifth worst team. By removing these teams, it was quite predicable that the hypothetical league would increase balance in regards to dominance and in-game equality.

It should be noted that while the Suncorp Super Netball league was a less competitive league in concentration, it was more balanced in regards to dominance and in-game evenness. Table 46 shows the differences between the two leagues.

Table 46: Meta Analysis of Suncorp Super Netball to Hypothetical League One

Measure	Suncorp Super Netball	Hypothetical League One	Difference
Lifetime Achievement (Range)	6.80	7.44	0.64
Lifetime Achievement (Std. Dev)	2.19	2.67	0.48
Blow-Out Index	33.04%	36.04%	3.00%
Average Margin	8.84	9.06	0.22

Hypothetical League One saw improvements in dominance and in-game evenness, however, these improvements were inferior to those found in the Suncorp Super Netball League. Furthermore, while concentration did improve, the improvement was so minor as to be inconsequential. The costs of having a trans-Tasman competition, such as the financial burdens on international travel on club and the apathy of TV spectators to trans-Tasman contest, outweigh the marginal improvements in competitive balance. Therefore, the conclusion is that the Suncorp Super Netball was a better option than Hypothetical League One, *ceteris paribus*.

8.3: Hypothetical League Two: The Suncorp Super Netball League without Adelaide

8.3.1: Introduction

The Adelaide Thunderbirds have struggled in the new Suncorp Super Netball League. After recording a solitary win in 2017, they went winless in 2018, and have a lifetime percentage of 76%. This is 20 percent points lower than the next worse team, the NSW Swifts. This is a regressive step from the Adelaide of the ANZ Championship, which was ranked fifth over the lifetime league, and made two premierships.

There is little benefit to anyone in having a low-performing team in a league as an outlier. The uncertainty of outcome for games are lower, which results in diminished fan interest and commercial attractiveness. Furthermore, players who become disillusioned with their team's poor performance are more likely to leave to other clubs, exacerbating the problem. Leagues can resolve these issues in a number of ways; the most common is the reverse order draft. In the Australian Football League, clubs that are particularly poor may be assigned a "priority pick", an additional first round pick unrelated to their position in the draft. This practice is controversial, as it provides some teams with an incentive to lose, and can therefore damage the integrity of the competition (Denham, 2011). In the Suncorp Super Netball league, there is no draft system, and players are recruited locally. Due to the relatively low salaries (the minimum salary is \$27,375, with the average rising to \$67,500) (AAP, 2016), many players take on study, work or family commitments that makes transferring clubs more difficult. Without a reverse order draft or a liquid free-agency market, there are few mechanisms available to improve the performance of Adelaide to an acceptable standard.

A hypothetical solution would be to remove Adelaide from the competition, and run a seven league team. While obviously this is a poor result for the netball fans in South Australia, this is a potential solution if the Suncorp Super Netball League wishes to maximise the competitive balance of their product. To test this theory, the 2017 and 2018 seasons were adapted to remove all games against Adelaide. As the league had a balanced roster, i.e. each team played every other team home and away, this adaption would not create a nested bias. Once the results of this new hypothetical league were created, four measures of competitive balance were applied, as follows:

- LAM (Dominance)
- HHI (Concentration)
- Blow-Out Index (In-Game Evenness)
- NAMSII (Concentration)

These four measures were chosen for their insightfulness, their compatibility with leagues of different sizes and their support within the literature. This section examines the results of each of these measures, and concludes with a summary of the findings.

Table 47: Lifetime Achievement of Suncorp Super Netball without Adelaide

Teams	SSN	SSN W/O ADL	Difference
Adelaide Thunderbirds	-4.54	N/A	N/A
Collingwood Magpies	-0.59	-1.63	-1.04
Giants Netball	2.01	1.83	-0.18
Melbourne Vixens	2.16	1.63	-0.53
New South Wales Swifts	-1.41	-1.63	-0.22
Queensland Firebirds	0.98	0.20	-0.78
Sunshine Coast Lightning	2.27	2.04	-0.23
West Coast Fever	-0.76	-1.22	-0.46

Table 48: Lifetime Achievement of Suncorp Super Netball with Adelaide

Summary	SSN	SSN W/O ADL	Difference
Min	-4.54	-1.63	2.91
Max	2.27	2.04	-0.23
Range	6.80	3.67	-3.13
Std. Dev	2.19	1.55	-0.63
Median	0.20	0.20	0.01

The findings from the LAM calculations are unsurprising. By removing Adelaide, the minimum value rises by 2.91 from -4.54 to -1.63. This reduces the range, which is the measure between the best and worst teams, as well as the standard deviation, which measures the variance of the mean. In regard to having a balanced competition, the closer to zero the better.

These findings show that a league without Adelaide would be more competitive in respect of dominance measures, as it removes the outlier from the competition.

8.3.2: Herfindahl-Hirschman Index

Table 49: nHHI table for Suncorp Super Netball to Hypothetical League Two

nHHI	SSN	SSN W/O ADL	Difference
2017	0.756	0.884	-0.128
2018	0.536	0.429	0.107
Average	0.646	0.656	-0.010

The results of the nHHI show that the new league, the one without Adelaide, is actually less balanced than the original league, albeit by a very small margin (-0.010). In 2017, the league without Adelaide was more competitively balanced, while in 2018 the original league was more even. The reasoning for this is unclear.

8.3.3: In-game statistics

Table 50: Comparative Statistics of Lifetime Achievement for Hypothetical League Two

In-Game Stats	SSN	SSN W/O ADL	Difference
Games	112	84	-28
Average Score	56.38	56.86	0.47
Average Margin	8.43	6.71	-1.72
Median Margin	7	5	-2
Number of Blowouts	36	20	-16
Blow-Out Index (%)	32.14%	23.81%	-8.33%
Standard Deviation	7.49	5.89	-1.59

As predicted, the uncertainty in-game is far higher in the new league than the original Suncorp Super Netball competition. The Blow-Out index is significantly lower, with only 23.81% of games ending with margins over ten goals. This new figure is lower than the ANZ Championship average, as well as every individual season bar the 2015 season. Assuming that fans are most interested in smaller margins, one might conclude that the reduction of average and median margins translates to a more competitive league. While the new games in the ANZ League may be more competitive, the fact that there was a significant drop in Blow-Out games would have had a far greater effect on reducing these values.

In summation, the exclusion of Adelaide Thunderbirds from the Suncorp Super Netball league would have improved the in-game competitive balance of the completion.

8.3.4: National Measure of Seasonal Imbalance (NAMSI)

Table 51: Comparison of NAMSI for Hypothetical League Two

	SSN	SSN W/O ADL	Difference
2017	0.87	0.82	-0.05
2018	0.84	0.58	-0.26
Average	0.85	0.79	-0.16

While there was little difference in the 2017 season, there is a five-fold difference in the subsequent year. The end result is that on average, the Suncorp Super Netball competition without Adelaide is significantly more balanced in terms of concentration. This contrasts with the previous findings from the HHI, which found the two leagues rather balanced. The most obvious reason for this is that the two calculations measure different aspects. The nHHI uses points attained as its basic input, while NAMSI uses standard deviations.

8.3.5: Conclusion

By using the four measures of competitive balance, it is possible to draw insights into the new league's evenness in terms of concentration, dominance and within-game. This analysis is not exhaustive, but suggests there is value in analysing the hypothetical Suncorp Super Netball without Adelaide.

Table 52: Meta analysis table for Suncorp Super Netball to Hypothetical League Two

Meta Analysis of Hypothetical League Two	Suncorp Super Netball	Hypothetical League Two
Lifetime Achievement		
Descriptive Statistics		
nHHI		
NAMSI		
Total		

The findings suggest that removing Adelaide from the Suncorp Super Netball League would result in a more balanced league with respect to dominance, concentration and in-game measures. Of the four measures used, only one calculation, the nHHI, did not align with this position, and the difference between the two leagues was a very small 0.01 points.

A league without Adelaide would make the competition more even, but there are additional costs to consider. A smaller league would mean less games being played, which in turn would reduce TV and gate revenues. One of the main goals Netball Australia has for the Suncorp Super Netball League is

to create a “fully-professional” competition, and this becomes more difficult if the number of teams and revenue is reduced. Another goal of Netball Australia is to improve participation, as well as the performance of the national team. By not having a professional presence in South Australia, it may be harder to attract young fans to play the game, and lower grass-roots participation would, over time, damage the quality of the national team. With the rise of women’s cricket, Australian Rules football and soccer, this could be a major concern. This is, however, purely speculation - the state of Tasmania does not have a professional AFL team and still produces many elite level footballers every year.

Were Adelaide to continue being significantly weaker than their rivals for a decade, in the same way the Canterbury Tactix were in the ANZ Championship, then there would be a stronger argument to remove them from the competition.

9: Conclusion

The research question posited by this thesis asked whether the Suncorp Super Netball League more competitively balanced than the ANZ Championship. This question was then divided into four hypotheses, which focussed on competitive balance in regards to concentration, dominance, in-game evenness, and geographically based inequality. Each hypothesis was tested using a varieties of measures, with the averages of both leagues compared against the other. It was considered important to use a variety of measures, as they allowed deeper insights and a more persuasive arguing of the research question.

The first hypothesis focussed on the differences in competitive balance in regards to concentration by using five different measures. They were the Herfindahl-Hirschman Index, Relative Standard Deviations, National Measure of Seasonal Imbalance, Concentration Ratios, and the Surprise Index. All bar the last measure found the Suncorp Super Netball League to have become a more unbalanced league in regards to concentration. This was surprising, as one of the original arguments for disbanding the ANZ Championship was a presumed lack of competitive balance.

The second hypothesis investigated competitive balance in regards to dominance. The measures used were lifetime achievement, adjusted churn, and Herfindahl-Hirschman Index. In this instance, all three measures found the Suncorp Super Netball competition to be more balanced. The reason for this was the Canterbury Tactix and to a lesser extent the Central Pulse were perennial underperformers. However, it should be noted that this hypothesis was the weakest, as dominance is a measure of success between seasons. As the Suncorp Super Netball League had only finished its second season, the paucity of data prevents firm conclusions from being drawn.

In-game evenness was tested by the third hypothesis using descriptive statistics, the Blow-Out index, and average of real to idealised percentage differences, an original calculation developed for this thesis. By applying the concepts of realised standard deviations to deviations of percentages, it was possible to procure deeper and more precise insights into in-game balance. In all three measures, it was found that the Suncorp Super Netball was more balanced than the ANZ Championship. A driving reason for this was, again, the underperformance of the Canterbury and Central teams. 39.41% of all Blow-Outs featured one of these teams as the loser.

The final hypothesis was divided into two parts. The first tested the belief that the Australian teams were superior to their New Zealand rivals in the ANZ Championship. Using descriptive statistics, league score averages, and the LAM, it was found that the Australian teams were in fact stronger. The only exception was the Waikato Bay of Plenty of Magic, which was the sole New Zealand championship winner and fourth on the lifetime achievement table. To test whether this result was influenced by the uneven scheduling, a hypothetical version of the ANZ Championship was reimagined, with each team playing each other an equal number of times. A number of tests were conducted on this new dataset, and they showed that the uneven schedule actually reduced the difference between the two countries. As the New Zealand teams played more games against their comparably weaker local teams, their lack of talent was masked. The overall conclusion from testing this hypothesis was that the Australian teams were stronger than their New Zealand counterparts.

After testing the four hypotheses, it was found that competitive balance improved for measure of dominance and in-game equality, but decreases for concentration. To help provide policy recommendations, two hypothetical leagues were created by removing the weakest teams from their respective competitions. A selection of competitive balance measures were applied to these new leagues, which were then compared to the original competition.

The first hypothetical league reimagined the ANZ Championship without the two lowest performing teams, the Canterbury Tactix and the Central Pulse. This change resulted in an increase in competitive balance regarding dominance and in-game evenness, although it had inconsequential effect on concentration. These modest improvements did not justify the additional costs of international travel and low viewership of trans-Tasman matches, and thus this policy alternative was not recommended.

Hypothetical League Two recreated the Suncorp Super Netball league without the Adelaide Thunderbirds. This new league was analysed using lifetime achievement, the Herfindahl-Hirschman Index, the Blow-Out index, and the National Measure of Seasonal Imbalance. All but the Herfindahl-Hirschman Index found the new league to be more competitively balanced, while the HHI measure

found only an inconsequential difference between the two leagues. Improvements in all three categories of competitive balance must be tempered by the lack of data. Forcing Adelaide to disband after only two poor seasons would be premature, and would leave Australia's fifth largest city without a team. Were Adelaide to continue to be uncompetitive in the future, possible options could be a merger, ideally from one of the eastern states which have two teams. Alternatively, the club could go under the administration of an established club in another code, such as the Adelaide Crows, an AFL team. The three expansion teams have similar affiliation deals, and these teams have been successful thus far. In the short term, Netball Australia should continue to support Adelaide's inclusion in the Suncorp Super Netball league.

This thesis examined whether the Suncorp Super Netball league was more competitively balanced than the ANZ Championship. Analysis found that the in-game evenness and dominance of the Super Netball League was indeed more balanced, while in contrast, concentration balance decreased. These results indicate that the balance of the Super Netball League did not increase across the board, which might lead one to wonder whether it was a wise replacement for the ANZ Championship. Although the answer to this question is beyond the scope of this thesis, its results do suggest that Suncorp Super Netball is the superior league. Netball Australia has now secured significant financial support from its sponsors, many players have seen their salaries triple, and the residents of the Sunshine Coast can now access world-class netball. These three outcomes are highly valued by Netball Australia, and they outweigh a drop in concentration related balance. For this reason, the decision to create the Suncorp Super League should be considered a success.

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11:Appendices

11.1: Appendix One: ANZ Championship Tables

2008 ANZ Championship season									V·T·E
	Team	Pld	W	D	L	GF	GA	G%	Pts
1	 Waikato/Bay of Plenty Magic	13	10	0	3	687	599	114.69	20
2	 New South Wales Swifts	13	10	0	3	727	652	111.50	20
3	 Adelaide Thunderbirds	13	9	0	4	652	577	113.00	18
4	 Melbourne Vixens	13	9	0	4	673	620	108.55	18
5	 Queensland Firebirds	13	7	0	6	693	645	107.44	14
6	 Southern Steel	13	7	0	6	617	616	100.16	14
7	 Northern Mystics	13	5	0	8	625	637	98.12	10
8	 Canterbury Tactix	13	5	0	8	607	678	89.53	10
9	 West Coast Fever	13	2	1	10	605	697	86.80	5
10	 Central Pulse	13	0	1	12	471	636	74.06	1
Updated: 2008-11-22									

2009 ANZ Championship ladder									V·T·E
Pos	Team	Pld	W	D	L	GF	GA	G%	Pts
1	 Melbourne Vixens	13	12	0	1	769	614	125.24	24
2	 Waikato Bay of Plenty Magic	13	11	0	2	673	562	119.75	22
3	 Adelaide Thunderbirds	13	10	0	3	698	579	120.55	20
4	 Southern Steel	13	8	0	5	662	645	102.64	16
5	 Queensland Firebirds	13	8	0	5	700	690	101.45	16
6	 Canterbury Tactix	13	5	0	8	639	662	96.53	10
7	 West Coast Fever	13	5	0	8	666	735	90.61	10
8	 Northern Mystics	13	3	0	10	642	727	88.31	6
9	 New South Wales Swifts	13	2	0	11	709	748	94.79	4
10	 Central Pulse	13	1	0	12	594	790	75.19	2
Updated on 7 July 2009 (Round 14, Game 5)									

2010 ANZ Championship ladder										V•T•E
Pos	Team	Pld	W	L	B	GF	GA	Diff	G%	Pts
1	New South Wales Swifts	13	13	0	1	704	570	+134	123.51	26
2	Adelaide Thunderbirds	13	9	4	1	681	586	+95	116.21	18
3	Waikato Bay of Plenty Magic	13	9	4	1	682	626	+56	108.95	18
4	Southern Steel	13	8	5	1	644	597	+47	107.87	16
5	Queensland Firebirds	13	7	6	1	717	629	+88	113.99	14
6	Northern Mystics	13	7	6	1	696	702	-6	99.15	14
7	Melbourne Vixens	13	6	7	1	651	680	-29	95.74	12
8	West Coast Fever	13	4	9	1	679	718	-39	94.57	8
9	Central Pulse	13	1	12	1	594	742	-148	80.05	2
10	Canterbury Tactix	13	1	12	1	571	769	-198	74.25	2
Updated on 22 June 2010 (Round 14, Game 5)										

2011 ANZ Championship ladder										V•T•E
Pos	Team	Pld	W	L	B	GF	GA	Diff	G%	Pts
1	Queensland Firebirds	13	13	0	1	758	587	+171	129.13	26
2	Waikato Bay of Plenty Magic	13	10	3	1	647	578	+69	111.94	20
3	New South Wales Swifts	13	9	4	1	677	606	+71	111.72	18
4	Northern Mystics	13	9	4	1	684	619	+65	110.50	18
5	Melbourne Vixens	13	8	5	1	664	610	+54	108.85	16
6	Adelaide Thunderbirds	13	5	8	1	662	737	-75	89.82	10
7	Southern Steel	13	4	9	1	533	594	-61	89.73	8
8	Central Pulse	13	3	10	1	599	683	-84	87.70	6
9	West Coast Fever	13	3	10	1	646	754	-108	85.68	6
10	Canterbury Tactix	13	1	12	1	621	723	-102	85.89	2
Updated on 2 May 2011 (Round 12, Game 5)										

2013 ANZ Championship ladder										V•T•E
Pos	Team	Pld	W	L	B	GF	GA	Diff	G%	Pts
1	Adelaide Thunderbirds	13	12	1	1	688	620	+68	111.0	24
2	Melbourne Vixens	13	9	4	1	692	589	+103	117.5	18
3	Waikato Bay of Plenty Magic	13	9	4	1	749	650	+99	115.2	18
4	Queensland Firebirds	13	9	4	1	793	691	+102	114.8	18
5	Central Pulse	13	8	5	1	736	706	+30	104.2	16
6	Southern Steel	13	6	7	1	812	790	+22	102.8	12
7	West Coast Fever	13	5	8	1	715	757	-42	94.5	10
8	New South Wales Swifts	13	4	9	1	652	672	-20	97.0	8
9	Canterbury Tactix	13	2	11	1	700	882	-182	79.4	4
10	Northern Mystics	13	1	12	1	699	879	-180	79.5	2
Updated on 25 June 2013 ↗ (Round 14, Game 5)										

2012 ANZ Championship ladder										V•T•E
Pos	Team	Pld	W	L	B	GF	GA	Diff	G%	Pts
1	Melbourne Vixens	13	10	3	1	645	569	+76	113.36	20
2	Northern Mystics	13	10	3	1	667	633	+34	105.37	20
3	Waikato Bay of Plenty Magic	13	9	4	1	699	594	+105	117.68	18
4	Adelaide Thunderbirds	13	9	4	1	670	589	+81	113.75	18
5	New South Wales Swifts	13	8	5	1	624	638	-14	97.81	16
6	Queensland Firebirds	13	7	6	1	686	640	+46	107.19	14
7	Central Pulse	13	5	8	1	585	626	-41	93.45	10
8	West Coast Fever	13	3	10	1	608	673	-65	90.34	6
9	Southern Steel	13	2	11	1	639	728	-89	87.77	4
10	Canterbury Tactix	13	2	11	1	634	767	-133	82.66	4
Updated on 2 July 2012 (Round 14, Game 5)										

2014 ANZ Championship ladder										V•T•E
Pos	Team	Pld	W	L	B	GF	GA	Diff	G%	Pts
1	Melbourne Vixens	13	9	4	1	746	592	+154	126.0	18
2	Queensland Firebirds	13	9	4	1	694	623	+71	111.4	18
3	New South Wales Swifts	13	9	4	1	707	652	+55	108.4	18
4	Waikato Bay of Plenty Magic	13	8	5	1	727	672	+55	108.2	16
5	Southern Steel	13	7	6	1	792	809	-17	97.9	14
6	Central Pulse	13	7	6	1	658	675	-17	97.5	14
7	Northern Mystics	13	6	7	1	706	752	-46	93.9	12
8	Adelaide Thunderbirds	13	5	8	1	663	706	-43	93.9	10
9	West Coast Fever	13	4	9	1	689	724	-35	95.2	8
10	Canterbury Tactix	13	1	12	1	694	871	-177	79.7	2
Updated on 22 June 2014 (Round 14, Game 5)										

2015 ANZ Championship (Conference)

Australian Conference									
Pos	Team	P	W	D	L	GF	GA	%	PTS
1	Queensland Firebirds (C)	13	11	1	1	708	608	116.4%	23
2	NSW Swifts	13	8	3	2	746	665	112.2%	19
3	West Coast Fever	13	9	1	3	744	701	106.1%	19
4	Melbourne Vixens	13	7	0	6	670	657	102.0%	14
5	Adelaide Thunderbirds	13	4	2	7	667	682	97.8%	10

New Zealand Conference									
Pos	Team	P	W	D	L	GF	GA	%	PTS
1	Northern Mystics	13	7	2	4	682	695	98.1%	16
2	Waikato Bay of Plenty Magic	13	5	1	7	643	663	97.0%	11
3	Southern Steel	13	3	2	8	759	775	97.9%	8
4	Central Pulse	13	3	2	8	620	656	94.5%	8
5	Mainland Tactix	13	1	0	12	664	801	82.9%	2

2016 ANZ Championship (Conference)

Australian Conference									
Pos	Team	W	L	D	GF	GA	GD	%	PTS
1	Queensland Firebirds	11	2	0	796	656	140	121.3	22
2	New South Wales Swifts	10	2	1	828	670	158	123.6	21
3	Melbourne Vixens	8	5	0	731	679	52	107.7	16
4	West Coast Fever	7	6	0	756	707	49	106.9	14
5	Adelaide Thunderbirds	2	11	0	660	775	-115	85.2	4

New Zealand Conference									
Pos	Team	W	L	D	GF	GA	GD	%	PTS
1	Southern Steel	11	0	2	852	732	120	116.4	24
2	Waikato Bay of Plenty Magic	6	7	0	665	755	-90	88.1	12
3	Northern Mystics	3	9	1	674	743	-69	90.7	7
4	Canterbury Tactix	2	10	1	708	825	-117	85.8	5
5	Central Pulse	2	10	1	676	804	-128	84.1	5

- Bye into Conference Finals.
- Participates in Finals Series.

11.2: Appendix Two: Suncorp Super Netball Tables

2017 Suncorp Super Netball Results:

#	Suncorp Super Netball Ladder							
	Team	W	D	L	GF	GA	%	PTS
1	Melbourne Vixens	11	1	2	874	774	118	23
2	Sunshine Coast Lightning	11	1	2	808	726	111	23
3	Giants Netball	10	0	4	773	728	106	20
4	Collingwood Magpies	9	0	5	770	730	106	18
5	Queensland Firebirds	7	1	6	778	756	103	15
6	NSW Swifts	3	1	10	726	792	92	7
7	West Coast Fever	2	0	12	670	779	86	4
8	Adelaide Thunderbirds	1	0	13	648	792	82	2

2018 Suncorp Super Netball ladder										V•T•E
Pos	Team	P	W	D	L	GF	GA	%	BP	PTS
1	Giants Netball	14	10	1	3	841	776	108.4	34	76
2	West Coast Fever	14	10	0	4	912	851	107.2	31	71
3	Queensland Firebirds	14	9	0	5	858	751	114.2	33	69
4	Sunshine Coast Lightning	14	8	1	5	809	752	107.6	35	69
5	Melbourne Vixens	14	8	0	6	855	814	105.0	27	59
6	New South Wales Swifts	14	6	1	7	814	816	99.8	22	48
7	Collingwood Magpies	14	3	1	10	801	858	93.4	25	39
8	Adelaide Thunderbirds	14	0	0	14	673	945	71.2	4	4
Last updated: 10 August 2018 — Source										

11.3: Appendix Three: Even Schedule League Tables

Master Table

Life Time Even Draw	Played	Won	Lost	Drawn	Points	Margin	Rank
Queensland Firebirds	81	61	19	1	123	756	1
Melbourne Vixens	81	58	23	0	116	698.5	2
New South Wales Swifts	81	54	24	3	111	561	3
Waikato Bay of Plenty Magic	81	50	31	0	100	496	4
Adelaide Thunderbirds	81	44	35	2	90	494.5	5
Southern Steel	81	39	40	2	80	291.5	6
West Coast Fever	81	35	46	0	70	381	7
Northern Mystics	81	30	49	2	62	179.5	8
Central Pulse	81	18	62	1	37	98	9
Canterbury Tactix	81	10	70	1	21	44	10

2008

2008 Even Draw	Played	Won	Lost	Drawn	Points	Margin	Rank
New South Wales Swifts	9	7	2	0	14	69.5	1
Waikato Bay of Plenty Magic	9	7	2	0	14	61	2
Melbourne Vixens	9	7	2	0	14	53.5	3
Adelaide Thunderbirds	9	6	3	0	12	73	4
Southern Steel	9	5	3	1	11	28	5
Queensland Firebirds	9	5	4	0	10	65	6
West Coast Fever	9	3	5	1	7	27	7
Canterbury Tactix	9	3	6	0	6	9	8
Northern Mystics	9	2	7	0	4	14	9
Central Pulse	9	0	9	0	0	0	10

2009

2009 Even Draw	Played	Won	Lost	Drawn	Points	Margin	Rank
Melbourne Vixens	9	8	1	0	16	126	1
Adelaide Thunderbirds	9	7	2	0	14	110	2
Waikato Bay of Plenty Magic	9	7	2	0	14	72.5	3
Queensland Firebirds	9	6	3	0	12	32	4
Southern Steel	9	5	4	0	10	39.5	5
West Coast Fever	9	4	5	0	8	30	6
Canterbury Tactix	9	4	5	0	8	26.5	7
New South Wales Swifts	9	2	7	0	4	21	8
Northern Mystics	9	1	8	0	2	2.5	9
Central Pulse	9	1	8	0	2	1	10

2010

2010 Even Draw	Played	Won	Lost	Drawn	Points	Margin	Rank
New South Wales Swifts	9	9	0	0	18	95.5	1
Adelaide Thunderbirds	9	6	3	0	12	103	2
Queensland Firebirds	9	6	3	0	12	93	3
Waikato Bay of Plenty Magic	9	6	3	0	12	48	4
Southern Steel	9	5	4	0	10	56	5
Melbourne Vixens	9	5	4	0	10	40.5	6
Northern Mystics	9	4	5	0	8	28	7
West Coast Fever	9	3	6	0	6	37	8
Canterbury Tactix	9	1	8	0	2	2	9
Central Pulse	9	0	9	0	0	0	10

2011

2011 Even Draw	Played	Won	Lost	Drawn	Points	Margin	Rank
Queensland Firebirds	9	9	0	0	18	140.5	1
New South Wales Swifts	9	8	1	0	16	53.5	2
Waikato Bay of Plenty Magic	9	7	2	0	14	55.5	3
Melbourne Vixens	9	5	4	0	10	73.5	4
Northern Mystics	9	5	4	0	10	46	5
Adelaide Thunderbirds	9	4	5	0	8	22	6
Southern Steel	9	3	6	0	6	10	7
West Coast Fever	9	2	7	0	4	9	8
Central Pulse	9	2	7	0	4	4.5	9
Canterbury Tactix	9	0	9	0	0	0	10

2012 Even Draw	Played	Won	Lost	Drawn	Points	Margin	Rank
Melbourne Vixens	9	7	2	0	14	61	1
Northern Mystics	9	7	2	0	14	45	2
Waikato Bay of Plenty Magic	9	6	3	0	12	112	3
Queensland Firebirds	9	5	3	1	11	66.5	4
Adelaide Thunderbirds	9	5	4	0	10	62.5	5
New South Wales Swifts	9	4	4	1	9	26	6
West Coast Fever	9	4	5	0	8	50	7
Central Pulse	9	3	6	0	6	18	8
Canterbury Tactix	9	1	8	0	2	3	9
Southern Steel	9	1	8	0	2	1	10

2013

2013 Even Draw	Played	Won	Lost	Drawn	Points	Margin	Rank
Adelaide Thunderbirds	9	8	1	0	16	63	1
Queensland Firebirds	9	7	2	0	14	82.5	2
Melbourne Vixens	9	6	3	0	12	88	3
Waikato Bay of Plenty Magic	9	6	3	0	12	76	4
West Coast Fever	9	5	4	0	10	57	5
Central Pulse	9	5	4	0	10	35.5	6
Southern Steel	9	4	5	0	8	36	7
New South Wales Swifts	9	3	6	0	6	39.5	8
Canterbury Tactix	9	1	8	0	2	3	9
Northern Mystics	9	0	9	0	0	0	10

2014

2014 Even Draw	Played	Won	Lost	Drawn	Points	Margin	Rank
Melbourne Vixens	9	7	2	0	14	110	1
Queensland Firebirds	9	6	3	0	12	77	2
New South Wales Swifts	9	6	3	0	12	62	3
Southern Steel	9	6	3	0	12	17.5	4
Waikato Bay of Plenty Magic	9	5	4	0	10	54	5
Central Pulse	9	4	5	0	8	24	6
Adelaide Thunderbirds	9	4	5	0	8	21	7
Northern Mystics	9	4	5	0	8	20	8
West Coast Fever	9	3	6	0	6	45	9
Canterbury Tactix	9	0	9	0	0	0	10

2015

2015 Even Draw	Played	Won	Lost	Drawn	Points	Margin	Rank
Queensland Firebirds	9	9	0	0	18	80.5	1
New South Wales Swifts	9	7	1	1	15	65.5	2
Melbourne Vixens	9	7	2	0	14	52.5	3
West Coast Fever	9	6	3	0	12	49.5	4
Northern Mystics	9	4	4	1	9	16.5	5
Adelaide Thunderbirds	9	2	5	2	6	23	6
Southern Steel	9	2	6	1	5	20.5	7
Central Pulse	9	2	6	1	5	8	8
Waikato Bay of Plenty Magic	9	2	7	0	4	7	9
Canterbury Tactix	9	1	8	0	2	0.5	10

2016

2016 Even Draw	Played	Won	Lost	Drawn	Points	Margin	Rank
Southern Steel	9	8	0	1	17	83	1
Queensland Firebirds	9	8	1	0	16	119	2
New South Wales Swifts	9	6	2	1	13	128.5	3
Melbourne Vixens	9	6	3	0	12	93.5	4
West Coast Fever	9	5	4	0	10	76.5	5
Waikato Bay of Plenty Magic	9	4	5	0	8	10	6
Northern Mystics	9	3	6	0	6	7.5	7
Adelaide Thunderbirds	9	2	7	0	4	17	8
Central Pulse	9	1	8	0	2	7	9
Canterbury Tactix	9	0	9	0	0	0	10

11.4: Appendix Four: ANZ Championship without Canterbury Tactix and Central Pulse Tables

2008

2008	Games Played	Wins	Draws	Average Winning Margin	Points	Winning Percentage
New South Wales Swifts	11	9	0	7.44	18	81.82%
Waikato Bay of Plenty Magic	9	6	0	8.50	12	66.67%
Adelaide	11	7	0	9.86	14	63.64%
Melbourne Vixens	11	7	0	7.43	14	63.64%
Queensland Firebirds	11	5	0	10.00	10	45.45%
Southern Steel	9	4	0	5.50	8	44.44%
Northern Mystics	9	2	0	4.00	4	22.22%
West Coast Fever	11	1	0	4.00	2	9.09%

2009

2009	Games Played	Wins	Draws	Average Winning Margin	Points	Winning Percentage
Melbourne Vixens	11	10	0	9.30	20	90.91%
Waikato Bay of Plenty Magic	9	7	0	11.43	14	77.78%
Adelaide	11	8	0	13.38	16	72.73%
Queensland Firebirds	11	7	0	3.57	14	63.64%
Southern Steel	9	4	0	6.75	8	44.44%
West Coast Fever	11	4	0	6.50	8	36.36%
New South Wales Swifts	11	1	0	4.00	2	9.09%
Northern Mystics	9	0	0	0.00	0	0.00%

2010

2010	Games Played	Wins	Draws	Average Winning Margin	Points	Winning Percentage
New South Wales Swifts	11	11	0	10.64	22	100.00%
Adelaide	11	7	0	10.14	14	63.64%
Waikato Bay of Plenty Magic	9	5	0	6.40	10	55.56%
Queensland Firebirds	11	5	0	12.00	10	45.45%
Southern Steel	9	4	0	6.75	8	44.44%
Melbourne Vixens	11	4	0	10.00	8	36.36%
Northern Mystics	9	3	0	5.00	6	33.33%
West Coast Fever	11	2	0	5.50	4	18.18%

2011

2011	Games Played	Wins	Draws	Average Winning Margin	Points	Winning Percentage
Queensland Firebirds	11	11	0	12.45	22	100.00%
Waikato Bay of Plenty Magic	9	6	0	9.67	12	66.67%
New South Wales Swifts	11	7	0	10.29	14	63.64%
Northern Mystics	9	5	0	6.80	10	55.56%
Melbourne Vixens	11	6	0	13.33	12	54.55%
Adelaide	11	3	0	8.33	6	27.27%
West Coast Fever	11	2	0	4.00	4	18.18%
Southern Steel	9	1	0	4.00	2	11.11%

2012

2012	Games Played	Wins	Draws	Average Winning Margin	Points	Winning Percentage
Adelaide	11	8	0	12.50	16	72.73%
Melbourne Vixens	11	8	0	7.13	16	72.73%
Northern Mystics	9	6	0	5.50	12	66.67%
New South Wales Swifts	11	7	0	5.14	14	63.64%
Waikato Bay of Plenty Magic	9	5	0	12.80	10	55.56%
Queensland Firebirds	11	5	0	8.00	10	45.45%
Southern Steel	9	1	0	2.00	2	11.11%
West Coast Fever	11	1	0	3.00	2	9.09%

2013

2013	Games Played	Wins	Draws	Average Winning Margin	Points	Winning Percentage
Adelaide	11	10	0	7.10	20	90.91%
Waikato Bay of Plenty Magic	9	6	0	13.00	12	66.67%
Melbourne Vixens	11	7	0	12.43	14	63.64%
Queensland Firebirds	11	7	0	12.43	14	63.64%
Southern Steel	9	4	0	9.25	8	44.44%
West Coast Fever	11	4	0	16.25	8	36.36%
New South Wales Swifts	11	3	0	14.00	6	27.27%
Northern Mystics	9	0	0	0.00	0	0.00%

2014

2014	Games Played	Wins	Draws	Average Winning Margin	Points	Winning Percentage
Waikato Bay of Plenty Magic	9	6	0	7.83	12	66.67%
Melbourne Vixens	11	7	0	18.43	14	63.64%
Queensland Firebirds	11	7	0	7.43	14	63.64%
New South Wales Swifts	11	7	0	7.29	14	63.64%
Southern Steel	9	5	0	4.20	10	55.56%
Northern Mystics	9	4	0	6.00	8	44.44%
Adelaide	11	3	0	5	6	27.27%
West Coast Fever	11	2	0	4.00	4	18.18%

2015

2015	Games Played	Wins	Draws	Average Winning Margin	Points	Winning Percentage
Queensland Firebirds	11	9	1	8.44	19	86.36%
New South Wales Swifts	11	7	2	10.43	16	72.73%
West Coast Fever	11	7	1	6.86	15	68.18%
Melbourne Vixens	11	5	0	7.40	10	45.45%
Waikato Bay of Plenty Magic	9	3	1	4.33	7	38.89%
Northern Mystics	9	3	1	4.00	7	38.89%
Adelaide	11	2	2	4.50	6	27.27%
Southern Steel	9	0	1	0.00	1	5.56%

2016

2016	Games Played	Wins	Draws	Average Winning Margin	Points	Winning Percentage
Southern Steel	9	8	1	9.13	17	94.44%
Queensland Firebirds	11	9	0	14.22	18	81.82%
New South Wales Swifts	11	8	0	13.75	16	72.73%
Melbourne Vixens	11	6	1	12.17	13	59.09%
West Coast Fever	11	5	0	11.20	10	45.45%
Waikato Bay of Plenty Magic	9	3	0	2.67	6	33.33%
Northern Mystics	9	1	0	3.00	2	11.11%
Adelaide	11	0	0	0.00	0	0.00%

11.5: Appendix 5: Suncorp Super Netball without Adelaide Tables

Master List

League 2018 w/o ADL	Played	Won	Lost	Drawn	For	Against	Percentage	Points
Collingwood Magpies	24	8	16	0	1330	1398	95%	2
Giants Netball	24	16	7	1	1485	1449	102%	17
Melbourne Vixens	24	15	7	2	1438	1356	106%	13
New South Wales Swifts	24	7	15	2	1315	1426	92%	11
Queensland Firebirds	24	12	10	1	1451	1389	104%	14
Sunshine Coast Lightning	24	16	6	2	1550	1456	106%	13
West Coast Fever	24	9	15	0	1446	1541	94%	16

2017

League 2017 w/o ADL	Played	Won	Lost	Drawn	For	Against	Percentage	Points
Collingwood Magpies	12	7	5	0	657	638	103%	14
Giants Netball	12	8	4	0	671	643	104%	16
Melbourne Vixens	12	9	2	1	728	640	114%	19
New South Wales Swifts	12	2	9	1	621	700	89%	5
Queensland Firebirds	12	5	5	1	670	669	100%	11
Sunshine Coast Lightning	12	10	1	1	698	634	110%	21
West Coast Fever	12	1	11	0	562	683	82%	2

2018

League 2018 w/o ADL	Played	Won	Lost	Drawn	For	Against	Percentage	Points
Collingwood Magpies	12	1	11	0	673	760	89%	2
Giants Netball	12	8	3	1	814	806	101%	17
Melbourne Vixens	12	6	5	1	710	716	99%	13
New South Wales Swifts	12	5	6	1	694	726	96%	11
Queensland Firebirds	12	7	5	0	781	720	108%	14
Sunshine Coast Lightning	12	6	5	1	852	822	104%	13
West Coast Fever	12	8	4	0	884	858	103%	16

12: Supplementary Table to Assist Reader

12.1: Summary of measures used in thesis

Table 53: Summary of measures used in thesis (Repeat of Table 1)

Name of Measure	Type of Measure	Summary	Strengths	Weaknesses
Herfindahl-Hirschman Index	Concentration	Measures clubs' share of wins	Simple to calculate. Well supported in the literature	Differences between sporting leagues and industry
Relative Standard Deviation	Concentration	Comparison between actual and ideal standard deviations	Can be used for leagues of different sizes	Upper bounds depend on team numbers
National Measure of Seasonal Imbalance	Concentration	Normalisation of Standard Deviations	Can be used for leagues of different sizes	Computationally complex
Concentration Ratios	Concentration	Measures subset against a broader group	Used in a variety of different sports	Weaker results when a league changes in size
Surprise Index	Concentration	Measure using upset victories in a given season	Can be widely used as a comparison tool	Laborious calculation process
Lifetime Achievement	Dominance	Measure of how many standard deviations away the W/L record is from the "ideal"	Can be used irrespective of sport, league, number of games played and time	Becomes more accurate with more data.
Adjusted Churn	Dominance	Measures the continuity of membership within a subset from year to year	Simple to calculate	Requires at least two years of data. Subset size is arbitrary
Herfindahl-Hirschman Index	Dominance	Measures club's proportional share of dominance	Simple to calculate. Well supported in the literature	Lack of seasons limits available data
Average Margins	In Game Competitiveness	Average margin of games	Simple to calculate	"The scoreboard didn't tell the whole story"
"Blow-Out" Index	In Game Competitiveness	% of games with margins > K	Can be widely used as a comparison tool	K value is arbitrary
Average of Real to Idealised Percentage Differences	In Game Competitiveness	The average of the absolute differences of actual percentages to the ideal percentage	Can be used for leagues of different sizes, and also to different sports	Original calculation developed for this thesis, and is thus unsupported in academic literature
League Score Average	Concentration	Summing the league rankings	Simple to calculate	Does not account for other factors

12.2: Summary of measures, their upper and lower bounds, as well as their directions

Table 54: Summary of measures, their upper and lower bounds, and directions

Name of Measure	Type of Measure	Direction	Lower Bound	Upper Bound
Herfindahl-Hirschman Index	Concentration	Low= More Balance High= Less Balance ↓	Various, but all less than 1	1
Relative Standard Deviation	Concentration	Low= More Balance High= Less Balance ↓	1	∞
National Measure of Seasonal Imbalance	Concentration	Low= More Balance High= Less Balance ↓	0	1
Concentration Ratios	Concentration	Low= More Balance High= Less Balance ↓	$\frac{D(2N - 2)}{W(2N - 1 - K)}$	1
Surprise Index	Concentration	Low= Less Balance High= More Balance ↓	0	1
Lifetime Achievement	Dominance	0 = Perfect Balance $\pm\infty$ = Less Balance ↕	0	$\pm\infty$
Time Series Associations	Dominance	Low= Less Balance High= More Balance ↑	0 or 0%	1, 100%, or N/2
Herfindahl-Hirschman Index	Dominance	Low= More Balance High= Less Balance ↓	1/N	1
Average Margins	In Game Competitiveness	Low= More Balance High= Less Balance ↓	0	∞
"Blow-Out" Index	In Game Competitiveness	Low= More Balance High= Less Balance ↓	0%	100%
Average of Real to Idealised Percentage Differences	In Game Competitiveness	Low= More Balance High= Less Balance ↓	0	∞
League Score Average	Dominance of a subset	Low= More Dominant High= Less Dominant ↑	For a ten team league = 0.38	For a ten team league = 1